Maintenance

Machine Maintenance

The maintenance schedule provides the maintenance task number along with a description of the task and the interval in hours. Each task should be performed at every interval indicated, not just the first instance. For the convenience of the end user, the following table is provided to cross-reference the hourly interval to alternate equivalent time units. Some of the alternate equivalency intervals are approximate.

⚠ WARNING

Prior to any maintenance or repair work being performed on the machine, standard Lockout/ Tagout procedures should be followed according to internal policy.

⚠ WARNING

Only qualified service technicians may perform maintenance and repair work on the machine.

NOTICE

If it is necessary to have repair work performed under warranty, it will help to have proof that regular maintenance work was performed and that the damage was not caused by insufficient or improper maintenance.

NOTICE

The following table presumes an eight-hour shift. If the end user deploys the equipment on a schedule that features either more or less shifts per day, then the alternate equivalent intervals must be adjusted accordingly.

NOTICE

The machine parts should be replaced by a trained technician who is familiar with the machine and good practices.

NOTICE

For more in-depth treatment of maintenance matters, BluePrint Automation offers a comprehensive Preventative Maintenance Program which is available from our service department.

HOURLY INTERVAL	ALTERNATE EQUIVALENT
8	Shift
16	Daily
96	Weekly
300	Triweekly
400	Monthly
800	Bimonthly
1250	Trimonthly
1600	Quarterly
2500	Semi-annually
5000	Annually
10 000	Biennially
20 000	Quadrennially

MAINTENANCE TASKS	ACTION	*Hours*
CLEAN THE MACHINE THOROUGHLY	CLEAN	8
CHECK FOR LOOSE PARTS	TIGHTEN	8
CHECK FOR MISSING, DAMAGED, OR WORN PARTS	REPLACE	8
CHECK FOR ANOMALOUS NOISES	ADJUST	8
CHECK THE TRANSPORT BELTS TRACKING AND WEAR	ADJUST	8
CHECK THE DRIVE BELTS FOR WEAR	ADJUST	8
CHECK TRANSPORT BELTS TENSION	ADJUST	96
CHECK DRIVE BELTS TENSION	ADJUST	96
CHECK BEARINGS AND SEALS FOR MISALIGNMENT	ADJUST	96
CLEAN ALL SENSOR HEADS AND REFLECTORS	CLEAN	96
CHECK GEARHEADS FOR OIL LEAKS	REPAIR/REPLACE	400
CHECK BEARINGS AND SHAFTS FOR EXCESS WEAR	ALIGN/REPLACE	1600
CHECK LUBRICATION IN GEARHEADS	LUBRICATE	2500
REPLACE THE COUNT BELT	REPLACE	2500
REPLACE THE DRIVE BELTS	REPLACE	5000
REPLACE THE TRANSPORT BELTS	REPLACE	5000
REPLACE THE FLANGE MOUNTED BEARINGS	REPLACE	5000
REPLACE THE RADIAL BALL BEARINGS	REPLACE	5000
REPLACE LUBRICANT IN THE GEARHEADS	REPLACE	10000
REPLACE THE GEARMOTORS SHAFT SEAL RINGS	REPLACE	10000
RELUBRICATE GEARMOTOR BEARINGS	LUBRICATE	20000

Recommended Products for Maintenance		
Application	**Category**	**Product name**
NORD gearhead roller bearings	Grease	Klüber Petamo® GHY 133 N
NORD Gearheads	Oil	Mobil SHC® 636
Threaded connections (non-stainless)	Adhesive	Henkel LOCTITE® 243
Housed bearings, o-ring seals	Grease	Henkel LOCTITE® LB 8104
Flat head screws in blind holes	Adhesive	Henkel LOCTITE® 242
Stainless steel screw threads	Oil	Henkel LOCTITE® LB 8060
Cylindrical fitting parts	Adhesive	Henkel LOCTITE® 609
Gearhead shaft bores	Grease	Henkel LOCTITE® LB 8014
General cleaning	Cleaner	ProChem® Indusol 2214

Daily Inspection

For safe and optimized machine performance, it is good practice to regularly inspect the machine for cleanliness, missing, worn, or damaged parts as well as the efficacy of all safety features.

Routine Cleaning

The following cleaning supplies are suggested:

- Clean damp cloth.
- Soft bristled brush.
- General water-based butyl cleaner/degreaser (such as ProChem® Indusol).

Remove all product and packaging materials from the machine and clean the interior surfaces, taking care to remove any accumulated product or adhesive residues.

Clean the operator’s HMI screen (with HMI off).

Routine Inspection

At a minimum, the following items should be checked for wear, damage, or corrosion etc.:

- Check sensors for efficacy.
- Check all moving parts for wear or damage.
- Check components that come in contact with the product or packaging material.
- Check door hardware integrity and safety switches efficacy and all other safety components for good functioning.

NOTICE

Any defects or needed repairs observed by the operator which are outside the operator’s purview are to be reported immediately to the maintenance department.

⚠ WARNING

Prior to any maintenance or repair work being performed on the machine, standard Lockout/Tagout procedures should be followed according to internal policy.

General Cleaning Recommendations

We recommend that you keep your belts' adhesive qualities intact by cleaning and conditioning belts when necessary. A clean belt will properly offer the friction qualities it should. Hygienic design of machinery, equipment and conveyor belts is the basis for safe food production. However, in the food industry this is not sufficient by itself. Manufacturers can only ensure that their products are reliably protected against pollution and contamination by regular and proper cleaning. Successful cleaning requires that the four main factors in the cleaning process are combined correctly:

- Mechanical/kinetic energy (cleaning method)
- Chemical energy (cleaning and disinfecting agent)
- Temperature
- Time

Cleaning methods in the food industry

A number of different cleaning methods are available in practice depending upon industrial sector, degree of contamination and level of equipment. Table below shows a survey of frequently used methods and average cleaning temperatures:

Method	COP * Cleaning/disinfecting		CIP * Cleaning/disinfecting
	Manual	Semi-automatic	Fully automatic
	• Sweeping • Scraping • Brushing • Scrubbing • Rinsing (hose) • Wiping • Vacuuming • Dipping	• Vacuuming • Spraying • Brushing + spraying • Dry vacuuming • Rinsing, spraying • Lathering, spraying	• Treatment in washing, rinsing and ultrasound machines • CIP (tanks, containers, pipes, etc.)
Temperature	Up to +40 °C/+104 °F	Up to +60 °C/+140 °F	Up to +80 °C/+176 °F

* CIP (Cleaning in Place), cleaning sealed systems
* COP (Cleaning on Place), surface cleaning (walls/ceilings/floors/conveyor belts/machinery, etc.)

When using the individual methods, please note the following points: 1 The use of brushes and cloths involves considerable hygiene risk (cross-contamination). 2 Subsequent spraying with or dipping in disinfectant is recommended. 3 The success of manual cleaning largely depends upon the subjective capabilities and care of the staff.

One method frequently used in practice to clean conveyor belts, especially in meat and fish processing, is foam cleaning or TFC cleaning (TFC = optimized foam cleaning).

Cleaning Agents

In the food industry, cleaning agents are selected on the basis of the type of soiling, corrosion resistance, parts or components being cleaned and the cleaning or disinfecting method. They may possess the following basic qualities:

- Completely and rapidly water soluble
- Wet all surfaces and materials to be cleaned well (not always necessary)
- Soak quickly and remove food residues (e.g. fats, proteins, carbohydrates, yeast, fruit pulp, etc.)
- Surface/material compatible
- Rinse off well

One single chemical does not usually possess all the qualities required. Consequently, in practice, combinations are used resulting in the following typical cleaning agents:

- Acid cleaners
- Neutral cleaners
- Alkaline cleaning agents
- Disinfectants

Examples of typical cleaning and disinfecting substances:

Acid cleaning agents	• Inorganic acids, e.g. phosphoric acid, nitric acid • Organic acids, e.g. acetic acid, citric acid • Solvents • Surfactants • Inhibitors
Neutral cleaning agents	• Phosphates • Surfactants • Peroxides • Complexing agents • Solvents
Alkaline cleaning agents	• Surfactants • Caustic soda solution/potash lye • Soda • Hypochlorites • Complexing agents
Disinfectants	• Peroxides • Peracetic acid • Quarternary ammonium compounds (QAV) • Hypochlorites • Aldehydes • Biguanides • Amphoterics • Alcohol • Chlorinedioxide ClO_2

Disinfectants reach a very wide range of micro-organisms, resistances of microbes do not occur. However, depending on the active matter some disinfectants do not kill specific germs (e.g. most of the surfactant based disinfectants avoid mold formation but are not suitable to kill high quantities of present mold). A change of the disinfectant makes sense if a micro species from an exterior source, which could not successfully be killed by the present disinfectant, enters the production plant. Frequently alkaline cleaning is used in the food processing environment because most of the present deposits are based on organic material. A sporadic acid cleaning is recommended to remove inorganic stains (e.g. water hardness deposits). However, areas being cleaned should always be rinsed thoroughly with water in between.

(Example W-A-W-S-W-D-W). (W = rinsing with water; A = alkaline cleaning; S = acid cleaning; D = disinfecting)

Recommended Cleaning Agents

- Soap and water: For all belt types; Nylon belts should be dried after washing
- Borax: Good for all belting
- Quaternary Ammonia: Useful for cleaning/ sanitizing food conveyor belts.

⚠ CAUTION

Combinations of agents may cause unpredictable damage or hazardous reactions. For all mentioned chemicals consult the manufacturer's safety data sheet (SDS) for proper safety precautions.

Action of cleaning agents and disinfectants

Agent	Action
Alkalis	Remove organic substances such as carbohydrates, fat and protein deposits.
Acids	Remove inorganic constituents (salts, calcium, lime-scale, beer stone, tartar, milk stone deposits).
Surface-active substances (surfactants)	Relieve surface tension of water. Penetrate and emulsify contaminants (e.g. fats, proteins, etc.)
Neutral cleaners	All-purpose cleaners with good cleaning action can be used on virtually any material (good fat and protein solvency).
Per compounds (Peracetic acid)	Oxygen-releasing compounds (oxidants) which act quickly. (Hydrogen peroxide) Suitable for chlorine-free disinfecting.
Active chlorine preparations	Primary potent disinfectants as well as for bleaching
Chlorinated alkaline compounds	Used to remove tenacious organic deposits such as protein- or carbohydrate-layer as well as bleaching.
Alcohol preparations	Fast-acting spray and surface disinfectants (destroying bacteria and other micro-organisms) are particularly suitable for surfaces susceptible to corrosion because they dry quickly.

Contamination	Cleaning agent base		Other components	
	Alkali's	Acids	Oxidants	Surfactants
Protein and protein deposits	Very good	Good	in special cases	Suitable
Oils and Fats	Suitable	Not Suitable	in special cases	Very good
Low-molecular carbohydrates	Very good	Very good	Not necessary	Not necessary
High-molecular carbohydrates	Suitable	Not necessary	Good	in special cases
Salts and minerals	Not Suitable	Very good	Not necessary	Not necessary

Cleaning machinery and conveyor belts

The following two cleaning methods are used in most areas of the food industry to clean machinery and conveyor belts. Certain parameters may vary depending upon industrial sector, severity of contamination and the material of the surface being cleaned.

- Cleaning temperature
- Contact time
- Cleaning agent
- Mechanical forces
- Disinfecting

General recommendation for cleaning conveyor belts

1. Preparation – switch off electric etc.
2. Gross solid removal

1. Pre-rinsing: With pre-rinsing, coarse dirt is rinsed off or detached with warm water (up to +60 °C) at low (max. 25 bar) pressure (high pressure causes increased ærosol formation –> recontamination, moreover, material exposed to high pressure is subject to excessive stress).
2. Rinsing off: In this stage, dirt previously detached or dissolved is rinsed off the belt with the aid of warm water
3. Cleaning: At the actual main cleaning stage, stubborn dirt on (up to +60 °C/+140 °F) and low pressure. It is particularly the belt (e.g. oils and fats) is dissolved with the important not to set the water pressure too high so aid of chemical cleaning agents. Cleaning that when rinsing off the conveyor belt neighbouring machinery, agents are generally applied as foam. In plant components, walls or floors are not contaminated again by practice, however, under certain circumstances splashes of material (cross-contamination) which has just been washed off the belt is also scrubbed manually.
4. Check cleaning result: Check all critical areas, e.g. visually or by ATP-measurement. Re-clean if necessary.
5. Disinfecting Disinfection is recommended in all hygiene relevant areas of a food processing plant such as production equipment, conveyor belts, packaging machines, facilities, floors, walls and all areas identified within the existing HACCP-system. It should always be taken into consideration that also surfaces without direct food contact could cause hygiene risks in regard to cross-contamination. Moreover, the conveyor belt must not be affected by cleaning agents and disinfectants. Therefore, note should be taken of the chemical resistance of the plastic, the directions for use and the dosage instructions of the chemicals used.
6. Final rinsing off with POTABLE water After cleaning, and if required, the conveyor belt is rinsed off with potable water using low pressure. (All residues from cleaning agents or disinfectants should be removed from the belt before it is used again).
7. Check: Disinfection result is checked using the appropriate method for the particular industrial sector (e.g. microbiological method: swab, contact plate).

It is important that all parts of the installation that come into contact with the belt are kept as clean as possible. Oil, grease, moisture, rust, dirt, traces of conveyed products, etc. on pulleys, rollers, slider bed and other parts of the installation in contact with the belt may cause operational and belt performance problems and will certainly shorten belt service life. The importance of belt cleanliness with regard to drive, proper tracking response and belt life cannot be overemphasized. Dirt on the conveying side of the belt may also lead to process breakdowns. Obviously, hygiene is of particular significance in the food sector where a number of special cleaning measures must be implemented.

Listed below a few general points on the cleaning of synthetic conveyor belts:

- Cleaning should, wherever and whenever possible, be carried out when the installation is at rest (safety aspect).
- In the case of light dirt deposits (dust, etc.), clean with a soft cloth; dry or moistened with cold or warm water.
- Oily, greasy soiling can be removed with hot water and a general, non-abrasive household detergent (low foaming types can aid the rinsing process).
- Spot cleaning can be performed via a damp rag application of a suitable solvent (see table).
- Heavy soiling can be removed by scrubbing with hot soapy water or washing with a mild solvent (see table).
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Unsuitable solvents/cleaners are:

- Aromatic compounds (benzene, toluene, xylene).
- Chlorinated hydrocarbons (trichloroethylene, tetrachloroethylene, tetrachlorohydrogen.
- Ketones (acetone, methyl ethyl ketone).
- Alkalis (Not for use on EPDM rubber).

- Ammonia (Not for use on NBR, Silicone or EPDM).
- Chlorine (Not for use on any belting).
- Kerosene (Not for use on PVC or EPDM).
- Live Steam (Not for use on Nylon belts).
- When working with inflammable and/or noxious chemicals, it is vital for you to observe all applicable safety precautions (refer to the corresponding safety data sheets for the chemicals to be employed).
- When cleaning with hot water or steam, take care not to exceed the maximum permitted temperature for the belt. The belt should be dried after being cleaned with water. Do not immerse belts in water or other fluids for long periods. This can result in irreversible dimensional changes (shrinkage), camber, color changes, degradation of the materials, layer separation or premature splice failure, etc.
- Where brushes are used for cleaning, use only those with soft bristles.

Notice: Inappropriate cleaning with high-pressure cleaning apparatus may damage the belt.

Polycarbonate cleaning

These cleaning recommendations apply to all polycarbonate sheet products. Periodic cleaning using correct procedures can help to prolong service life.

Note: Never use products made for cleaning glass (windows) or made with alcohol.

Cleaning Procedure for Small Areas

1. Gently wash the polycarbonate sheet with a solution of mild soap and lukewarm water, using a soft, grid-free cloth or sponge to loosen any dirt or grime.
2. Fresh paint splashes, grease and smeared glazing compounds can be removed easily before drying by rubbing lightly with a soft cloth using petroleum ether (BP65), hexane or heptane. Afterwards, wash the sheet using mild soap and lukewarm water.
3. Scratches and minor abrasions can be minimized by using a mild automobile polish. We suggest that a test be made on a small area of the sheet with the polish selected and that the polish manufacturer's instructions be followed, prior to using the polish on the entire sheet.
4. Finally, thoroughly rinse with clean water to remove any cleaner residue and dry the surface with a soft cloth to prevent water spotting.

Cleaning Procedure for Large Areas

1. Clean the surface using a high-pressure water cleaner (max. 100 bar or 1,450 psi) and/or a steam cleaner. We suggest that a test be made on a small area, prior to cleaning the entire sheet.
2. Use of additives to the water and/or steam should be avoided.

Other Important Instructions for All Polycarbonates:

- Never use abrasive or highly alkaline cleaner.
- Never use aromatic or halogenated solvents like toluene, benzene, gasoline, acetone or carbon tetrachloride on polycarbonate materials.
- Use of incompatible cleaning materials with polycarbonate (Lexan) sheet can cause structural and/or surface damage.
- Contact with harsh solvents such as methylethylketone (MEK) or hydrochloric acid can result in surface degradation and possible crazing of Polycarbonate.
- Never scrub with brushes, steel wool or other abrasive materials.
- Never use squeegees, razor blades or other sharp instruments to remove deposits or spots.
- Do not clean polycarbonate in direct sunlight or at high temperatures as this can lead to staining.

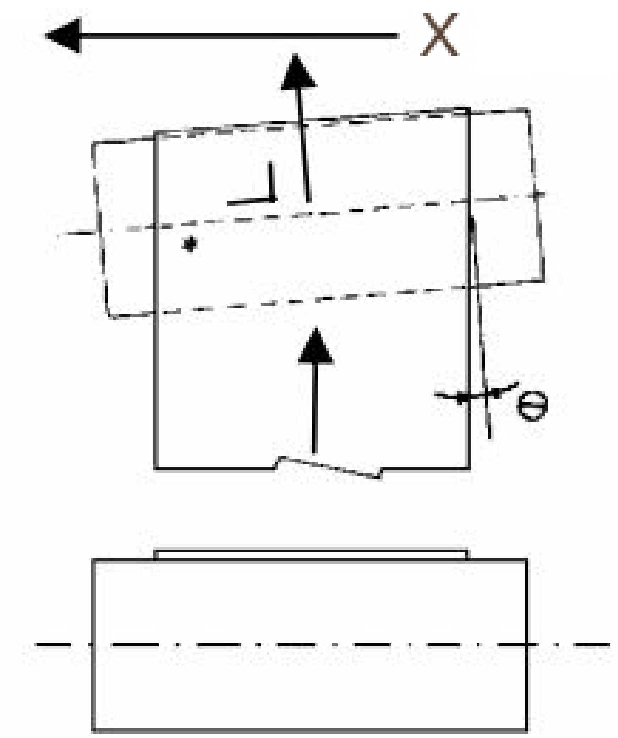
Conveyor Belt Maintenance

A belt running against machine parts because of mistracking results in fraying edges, shorter belt life, and possibly other damage to drive components. The particles from deteriorating belts can eventually contaminate the product. This can be prevented by proper belt tensioning and tracking.

Check conveyor belts for wear, fraying, tracking and tension. Check lacing or other joining technologies for integrity. Replace any unserviceable belts.

Conveyor Belt Tracking

When a pulley is not placed square to the belt, it will result in the situation as shown below. The belt will move to the left (**X**).



The problem is corrected by restoring perpendicularity of the pulley to the belt.

Conveyor Belt Tension

Belt tension must be great enough to prevent slippage between the drive pulley and belt.

Slippage will cause excessive wear to both drive pulley lagging and the belt.

Proper minimum belt tension is the tension required so that a given belt conveyor or belt elevator system will operate properly in its environment.

Operate the belt conveyor system under the most adverse conditions it will encounter (i.e. under load), and observe closely to see if there is any slippage between the drive pulley and the belt. If there is, more tension is required.

Do not over tension, minimum belt tension is just enough, not too much.

Conveyor Belt Installation

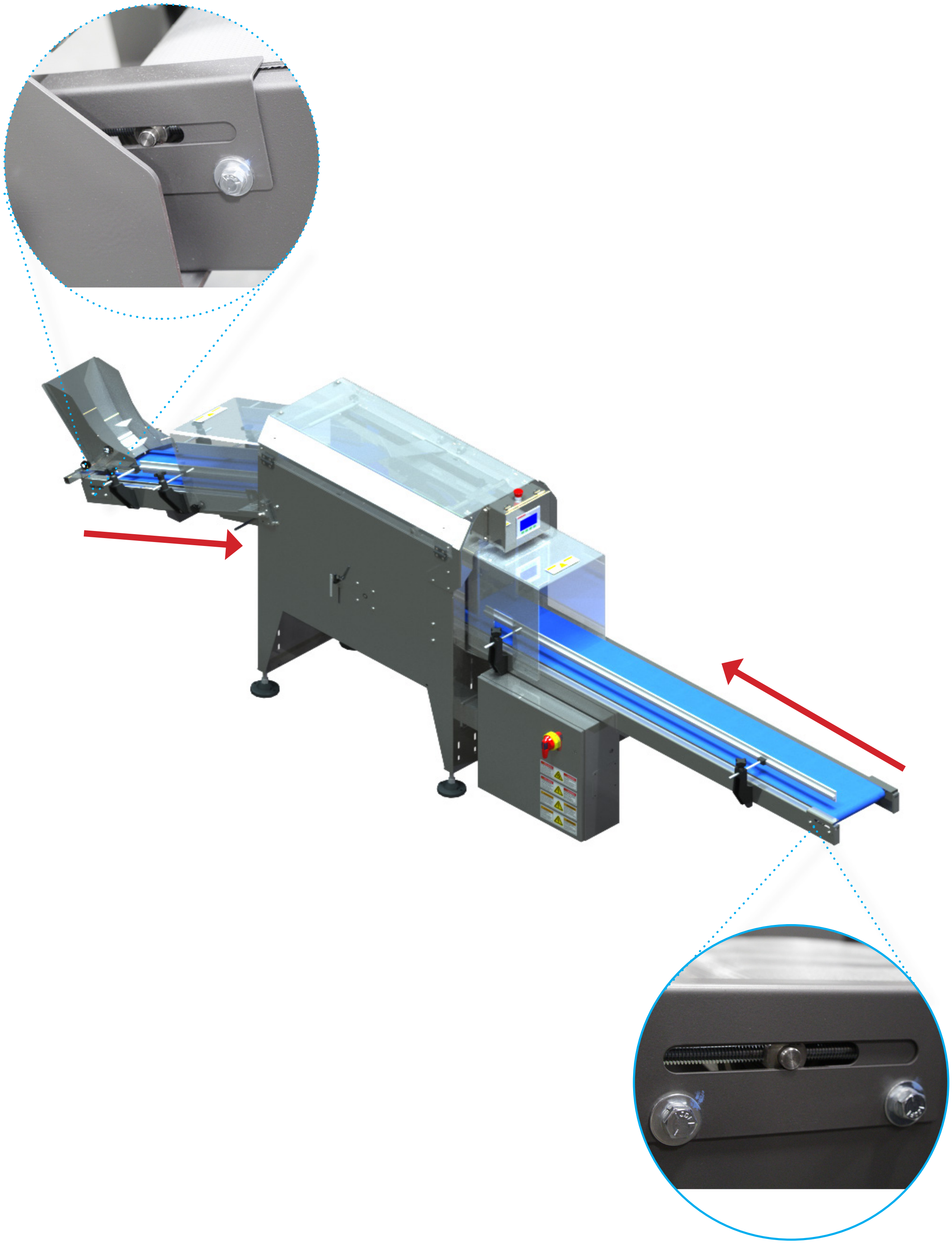
1. Do not pry or force the belt onto the pulleys.
2. Reduce the pulley's center distance so that the belt can be easily installed.
3. Place the belt on the pulleys; do not pry or force the belt.
4. Increase the belt tension until the belt feels snug or taut. Avoid over tensioning the belt.
5. Start the drive under the most severe load conditions. This may be either the motor starting torque or during the work cycle. A belt that is too loose will jump teeth at this time.
6. Tighten the belt gradually until a satisfactory operating condition is achieved.

Infeed and Exit Belts Maintenance

To adjust or replace the infeed or exit conveyor belts you must act on the various belt tensioners.

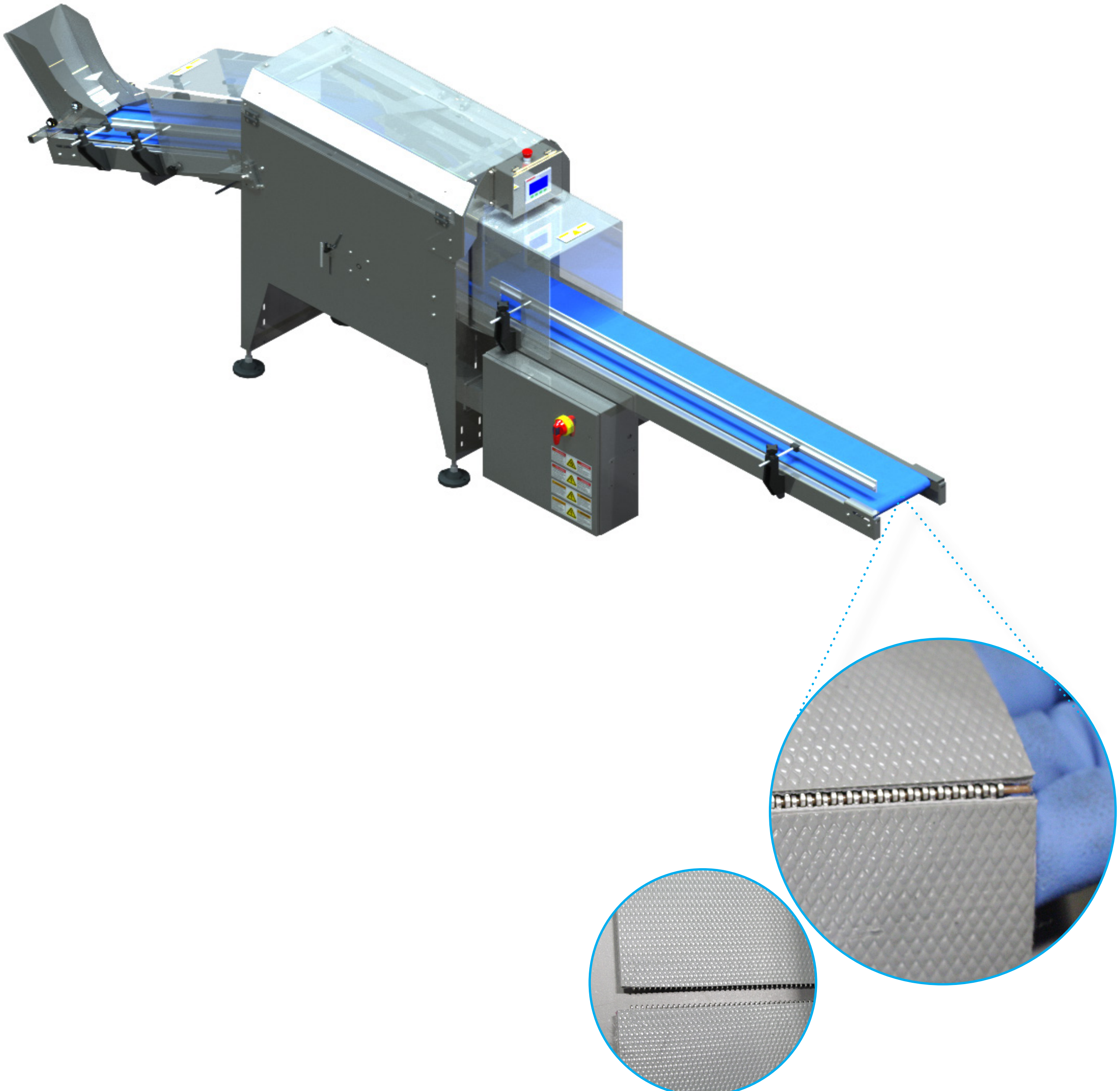
1

Loosen the two tension screws at the end of the shaft; there are two sets, one set on the infeed belt located under the infeed chute and the other set is located at the end of the exit belt.



2

Adjusting the screws will move the shaft back in toward base of machine as the red arrows indicate above, allowing the conveyor belt to loosen and have slack.



3

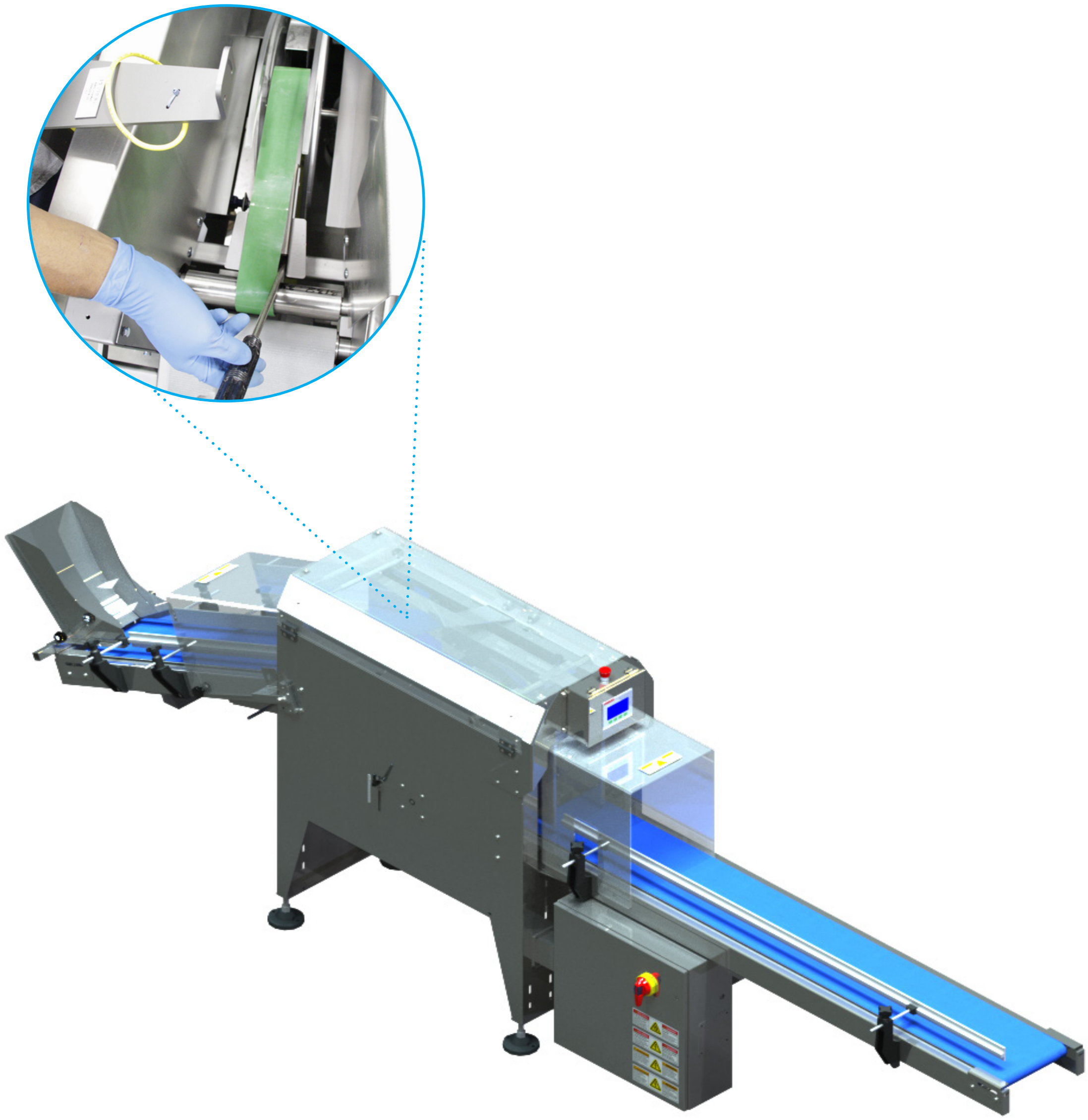
Locate where the belt binds together and remove the joining pin; allowing the ends to separate. Reassemble and apply tension via the tensioners.

Count Belt Maintenance

To adjust or replace the count conveyor belt, follow the steps indicated below.

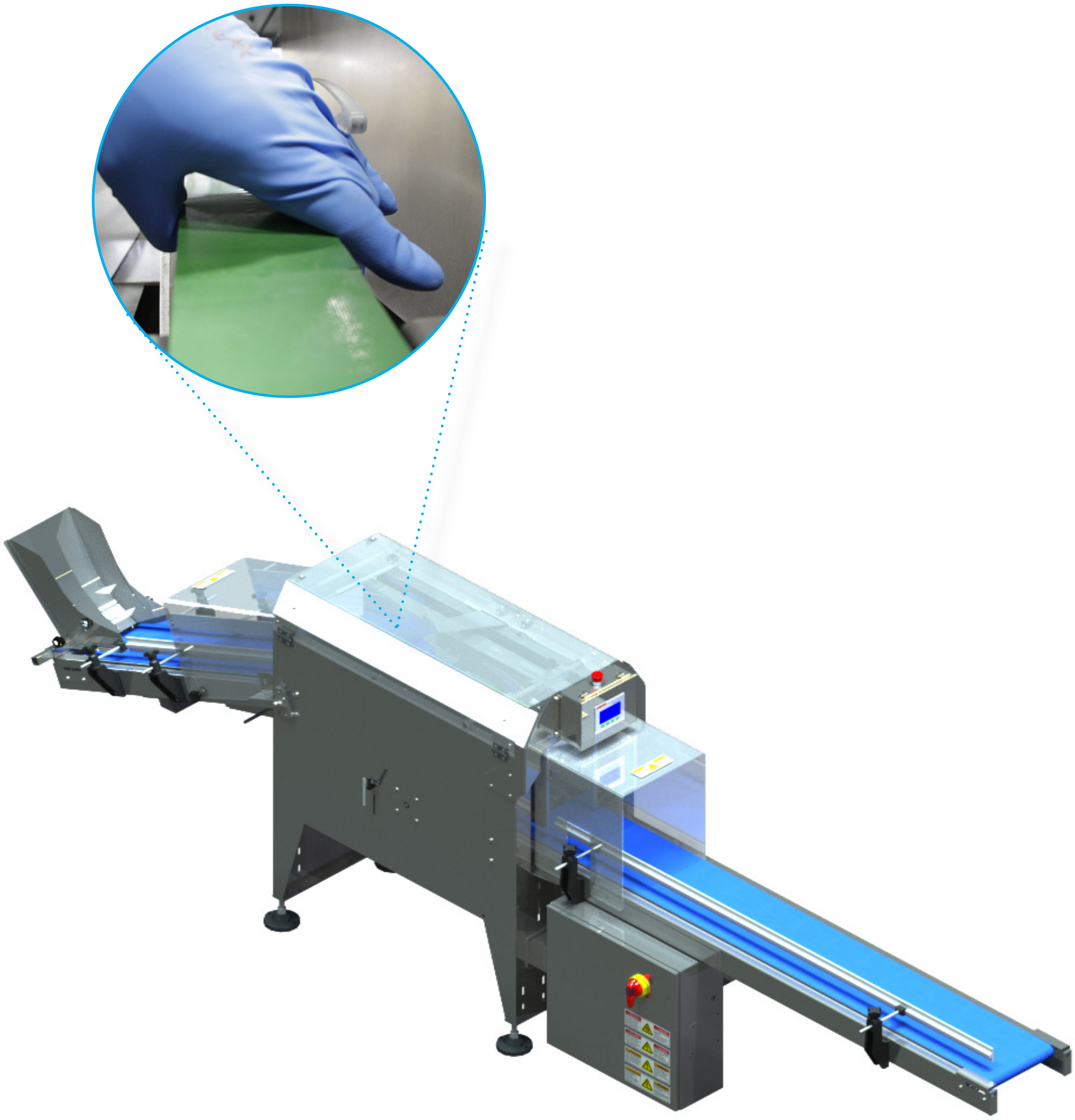
1

With the belt installed, use a 5/32” hexagon key, loosen the four 1/4’’ hexagon socket countersunk screws as shown above.



2

Use a lever to adjust the weldment; applying pressure with the lever will slide the weldment back and forth. Make sure to adjust both sides of the weldment equally, maintaining parallelism while at the same time, making sure to keep the infeed belt level with the package guide.



3

Make sure to go by the one finger rule; you should be able to put one finger in between the conveyor belt and the weldment. Once this is confirmed and infeed belt is adjusted to the desired tension, tighten the four 1/4” hexagon countersunk head screws with a 5/32” hexagon key.

Synchronous Belt Drives Maintenance

Check for damage, fraying, or wear, replace any unserviceable belts. Check tension and alignment per paragraphs below.

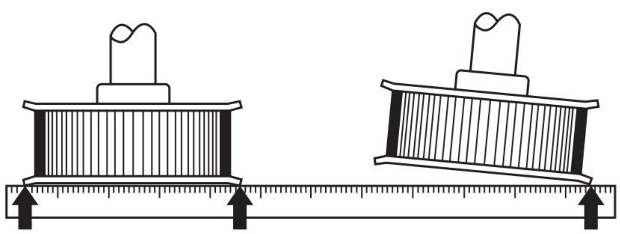
Belt Tension

Synchronous belt drives generally require little if any retensioning when used in accordance with proper design procedures. A small amount of belt tension decay can be expected within the first several hours of operation (running-in period). After this time, the belt tension should remain relatively stable.

Pulley alignment

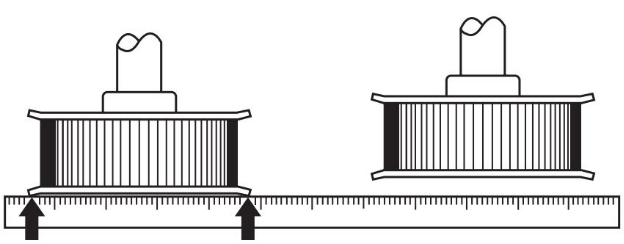
Drive misalignment is one of the most common sources of drive performance problems. Misaligned drives can exhibit symptoms such as high belt tracking forces, uneven belt tooth wear, high noise levels, and tensile cord failure. The two primary types of drive misalignment are angular and parallel. Discussion about each of these types are as follows:

Angular misalignment results when the drive shafts are not parallel. As a result, the belt tensile cords are not loaded evenly, resulting in uneven tooth/land pressure and wear. The edge cords on the high tension side are often overloaded which may cause an edge cord failure that propagates across the entire belt width.



Angular misalignment often results in high belt-tracking forces as well which cause accelerated belt edge wear, sometimes leading to flange failure or belts tracking off of the pulleys.

Parallel misalignment results from pulleys being



mounted out of line from each other. Parallel misalignment is generally more of a concern with V-type belts than with synchronous belts because V-type belts run in grooves and are unable to free float on the pulleys.

Synchronous belts will generally free float on the pulleys and essentially self-align themselves as they run. This self-aligning can occur as long as the pulleys have sufficient groove face width beyond the width of the belts. If not, the belts can become trapped between opposite pulley flanges causing serious performance problems. Parallel misalignment is not generally a significant concern with synchronous drives as long as the belts do not become trapped or pinched between opposite flanges.

Allowable Misalignment: In order to maximize performance and reliability, synchronous drives should be aligned closely. This is not, however, always a simple task in a production environment. The maximum allowable misalignment, angular and parallel combined, is 1/4°.

Angular misalignment is not always easy to measure or quantify. It is sometimes helpful to use the observed tracking characteristics of a belt, to make a judgment as to the system's relative alignment. Neutral tracking "S" and "Z" synchronous belts generally tend to track "down hill" or to a state of lower tension or shorter center distance when angularly misaligned. This may not always hold true since neutral tracking belts naturally tend to ride lightly against either one flange or the other due to numerous factors discussed in the section on belt tracking. This tendency will generally hold true with belts that track hard against a flange. In those cases, the shafts will require adjustment to correct the problem.

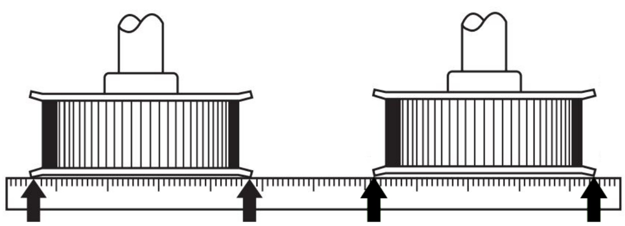
Parallel misalignment is not often found to be a problem in synchronous belt drives. If clearance is always observable between the belt and all flanges on one side, then parallel misalignment should not be a concern.

Belt Replacement

During the belt installation process, it is very important that the belt be fully seated in the pulley grooves before applying final tension. Serpentine drives with multiple pulleys and drives with large pulleys are particularly vulnerable to belt tensioning problems resulting from the belt teeth being only partially engaged in the pulleys during installation. In order to prevent these problems, the belt installation tension should be evenly distributed to all belt spans by rotating the system by hand. After confirming that belt teeth are properly engaged in the pulley grooves, belt tension should be rechecked and verified. Failure to do this may result in an undertensioned condition with the potential for belt ratcheting.

Belt Removal

1. Loosen and remove all screws from the bushing.
2. Put the screws in the threaded holes in the outer piece of the assembly.
3. Tighten the screws equally to force the mating part loose from bushing.
4. After replacing the pulley, the shafts and pulley must be properly aligned to ensure proper wear and extending belt life.
5. Place a straight edge against the pulleys.
6. Make certain the straight edge touches all four points of the pulleys. The straight edge should be at the widest possible distance on the pulleys.

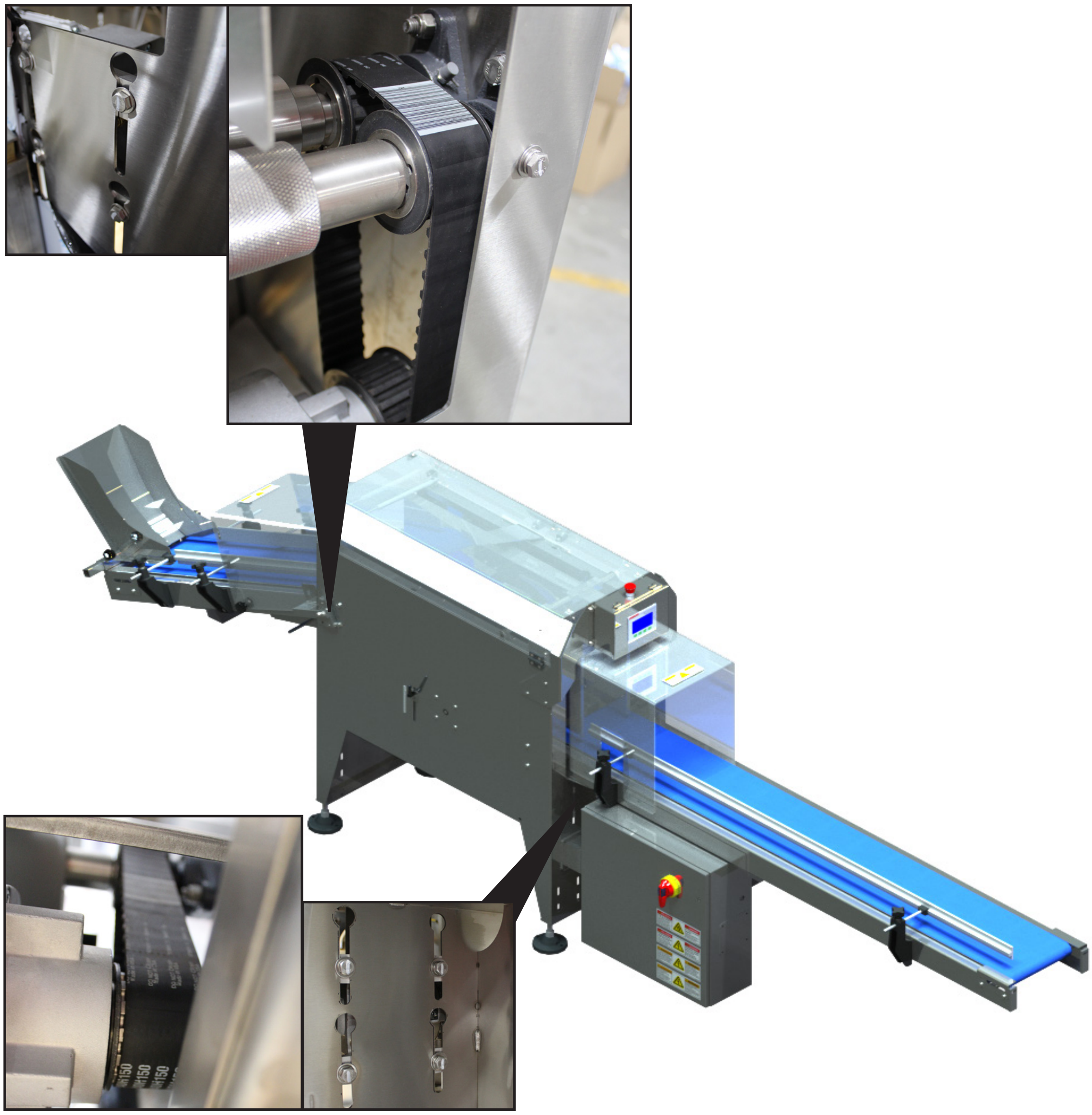


Belt Installation

1. Do not pry or force the belt on the pulleys.
2. Do one of the following:
3. Remove the pulley's flange
4. Reduce the pulleys' center distance so the belt can be easily installed
5. Place the belt on the pulleys. Do not pry or force the belt.
6. Increase the belt tension until the belt feels snug or taut. Avoid over tensioning the belt.
7. Start the drive under the most severe load conditions. This may be either the motor starting torque or during the work cycle. A belt that is too loose will jump teeth at this time.
8. Tighten the belt gradually until a satisfactory operating condition is achieved.

Synchronous Drive Belts Maintenance

To adjust or replace the synchronous drive belts, follow the steps indicated below.



1

Loosen the four hexagon head cap screws that mount the motor to the machine housing just enough to allow the motor to slide vertically within the slots. Now the motor can be slid up in the slots thereby loosening the belt. Tighten the screws to keep the motor from dropping while the belt is exchanged (if necessary). To adjust the tension, loosen the screws allowing the weight of the motor to apply downward force thus tensioning the belt. Apply some additional downward force to the motor until optimum tension is obtained - approximately 3-6 mm of deflection. Retighten the screws.

Rolling Bearing Maintenance

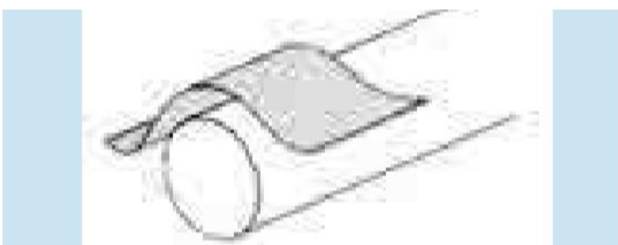
Lubrication

BluePrint Automation ball and roller bearing installations are considered lubricated for life and are therefore not furnished with grease fittings. Rather, we advise changing them out at their expected mean time to failure (*MTTF)*.

Replacement

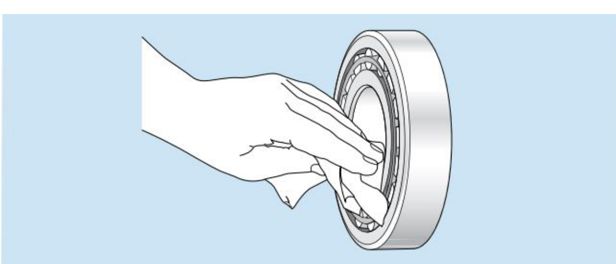
When replacing bearings, use appropriate tools (pullers, impact sleeves, impact rings, mandrels, lead-in bushings, presses, etc.). Do not pry bearings or apply uneven forces.

1. Make sure to burr the shafts and bores with an emery cloth or fine file.



2. Wipe the shaft with a clean cloth. New bearings do not need to be cleaned.

3. Wipe residual preservative from the bore of the bearing

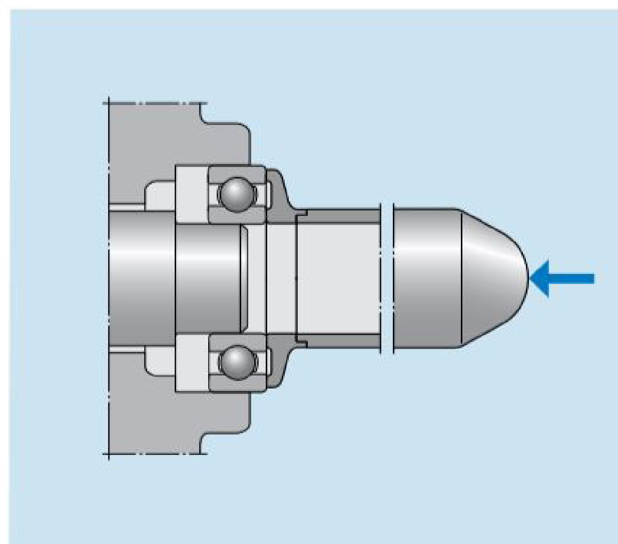


On mounting of non-separable bearings, the mounting forces must always be applied to the ring which will have the tight fit and therefore is the first to be mounted. Forces applied to the ring with the loose fit would be transmitted by the rolling elements, thus damaging raceways and rolling elements.

Mounting of separable bearings is easier, since the two rings can be mounted separately. In order to avoid score marks during assembly, slightly rotate the parts.

The use of a mounting disc is advisable when mounting simultaneously to shaft and housing assembly so that mounting forces are applied uniformly to inner and outer bearing races.

If a non-separable bearing is to be pressed onto the shaft and into the housing bore at the same time, the mounting force has to be applied equally to both rings at the same time and the abutment surfaces of the mounting tool must lie in the same plane. In this case a bearing fitting tool should be used, where an impact ring abuts the side faces of the inner and outer rings and the sleeve enables the mounting forces to be applied centrally.



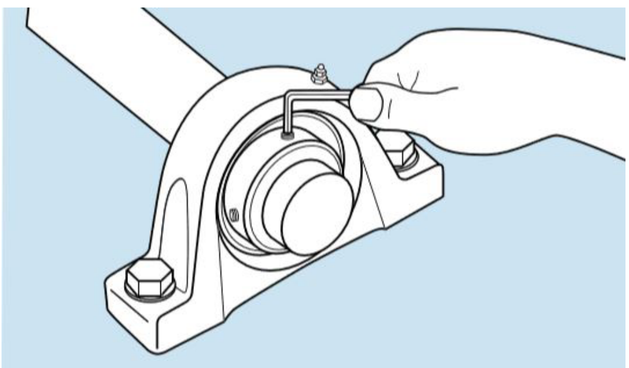
Housed Bearing Maintenance

Lubrication

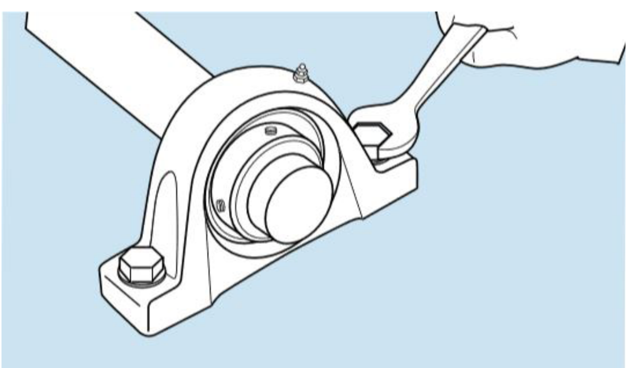
Generally speaking, pillow block and flange mounted bearing units are designed to operate without relubrication under normal speed and operating conditions. All ball bearing units are sealed at both sides with rubbing contact seals and are filled with a special long life grease of NLGI consistency 2. The grease has good corrosion inhibiting properties and is suitable for operating temperatures between –4 °F and +248 °F.

Some bearing units may be equipped with a grease fitting that allows the bearing to be relubricated in service. When relubricating, care must be taken to use greases that are compatible with the original grease. We suggest a medium temperature, lithium calcium base, NLGI 2 grease having a base oil with a viscosity of 900 SUS (200 mm²/s) at +100 °F (+40 °C). When a unit is being relubricated, avoid excessive pressure, which may cause damage to the bearing seals.

Dismounting instructions

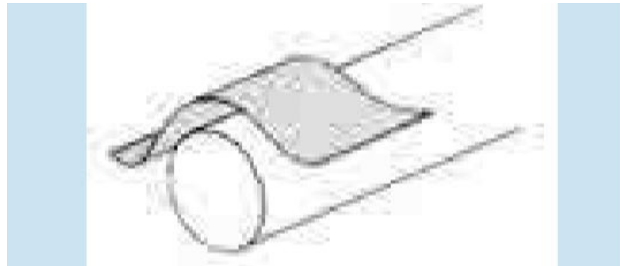


1. Loosen setscrews.
2. Unbolt the housing from its support. The complete bearing unit can then be removed from the shaft. It will be necessary to relieve the bearing load when removing the unit.



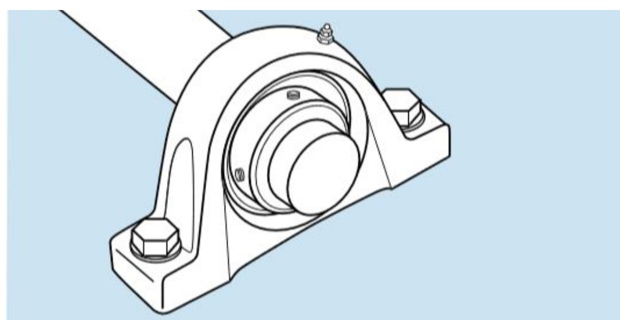
Mounting instructions

1. Remove any burrs or rust on the shaft with an emery cloth or a fine file.



2. Wipe the shaft with a clean cloth.

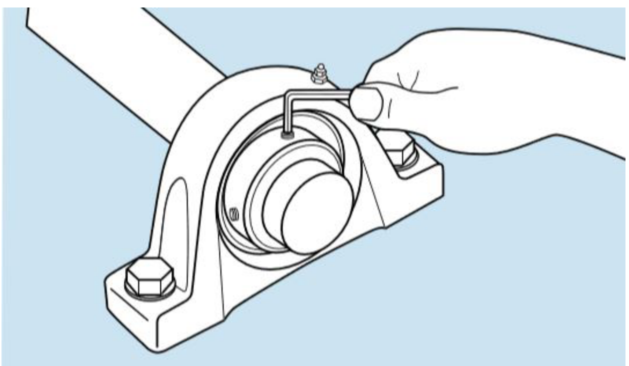
3. Check the shaft diameter
   Recommended shaft tolerances:
   Shaft diameter tolerance Up to 1 15/16” nominal to (49.2 mm) -0.0005” (-0.013 mm)
   2” to 4” Nominal to (50.8 to 101.6 mm) -0.0010” (-0.025 mm)
4. Slide the bearing and housing onto the shaft and position. For eccentric lock-type units, leave the collar loose on the shaft.



5. Clean the base of pillow block and the support surface on which it rests. Be sure the supporting surface is flat. If the pillow block elevation must be adjusted by shims, the shims *must* extend the full length and width of the support surface. Bolt the pillow block securely to the support. With flanged housings, clean the flange and support surface. Be sure the support surface is flat. Bolt the flanged housing securely to the support.

1. Tighten each setscrew alternately with proper hexagon head socket wrench until they stop turning and the hexagon head socket wrench starts to spring. The spring of the hexagon head socket wrench can be easily seen and felt when the extension is used When both setscrews are tightened on the shaft, the bearing is firmly seated. This completes the procedure for mounting the setscrew lock collars..See torque values.

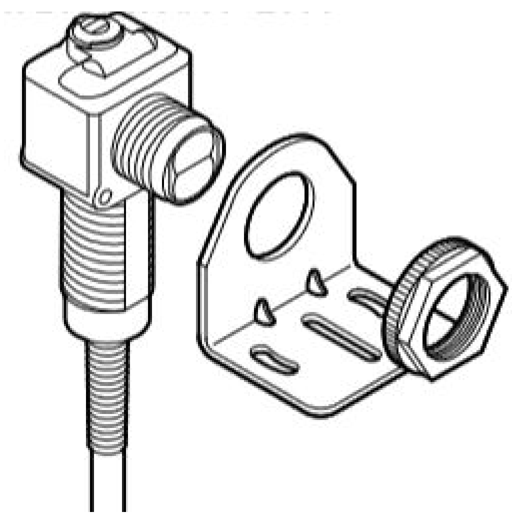
Torque Table	
Setscrew size	**Torque in·lb (Nm)**
#10-32 (M5)	36 (4.0)
1/4-28 (M6)	87 (9.8)
5/16-24 (M8)	165 (18.6)
3/8-24 (M10)	290 (32.8)
7/16-20 (M12)	430 (48.6)
1/2-20 (M14)	620 (70.1)

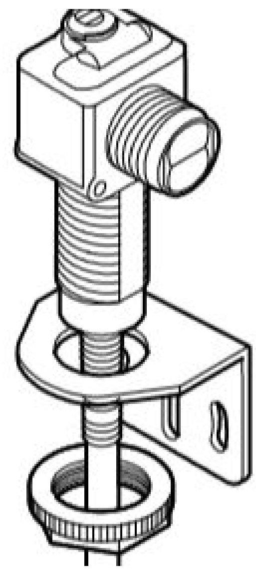


Photoelectric Sensors Maintenance

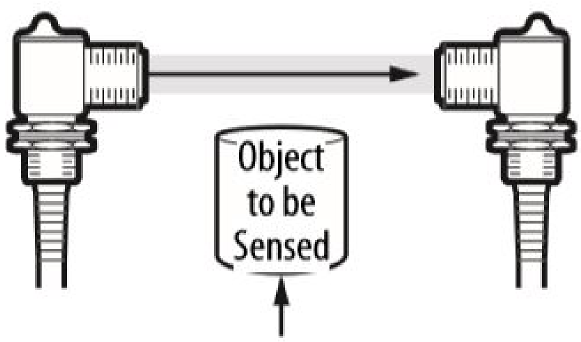
Various photoelectric sensors are used for product or position detection.

1. The sensors are typically held in place by a jam nut.

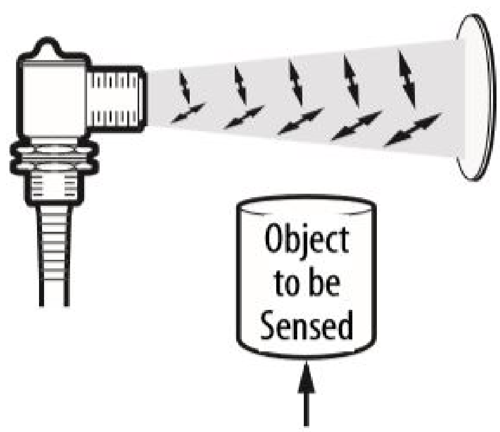




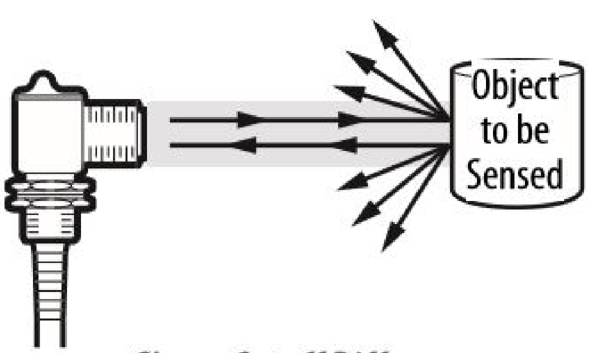
2. The sensor must not have any obstructions or debris in front of it.
3. Keep sensor heads, lenses, and reflectors clean.
4. The sensors must be adjusted according to the manufacturer specifications as sensing distances vary by make and model.
5. The following sensor types are typically deployed on Blueprint Automation equipment.



Transmitted Beam



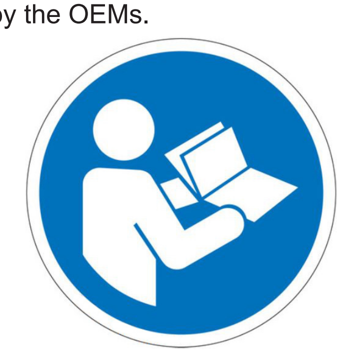
Polarized Retroreflective



Sharp Cutoff Diffuse

Sensor Replacement

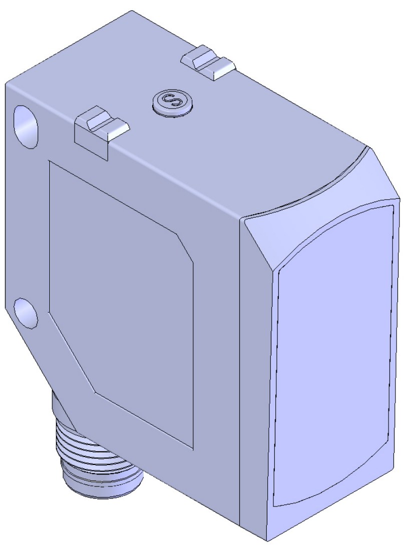
When replacing defective sensors it is advisable to consult the instruction bulletins furnished with the new sensors by the OEMs.



Clean The Laser Sensors

Build-up of dirt on the sensor's lenses may affect the efficacy of the devices. It is therefore recommended to perform maintenance regularly. The laser sensors are EN/IEC 60825-1 Class 1 laser products, that are considered safe under reasonably foreseeable conditions of operation, including the use of optical instruments for intrabeam viewing.

It is recommended to use compressed air to blow dust away. Clean with a dry or a damp cloth using a mild detergent if necessary. Do not use any abrasive or solvent cleaners.



Enclosure is rated IP67 according to IEC/EN 60529

Sensor Replacement

When replacing defective sensors it is advisable to consult the instruction bulletins furnished with the new sensors by the OEMs.

Asynchronous Gearmotor Maintenance

Asynchronous (or induction) gearmotors should be inspected regularly for leaks, cracks, and external damage.

Check for anomalous running noises; gearheads exhibiting unusual running noises and/or vibrations could indicate damage to the gearhead. In this case the gearhead should be shut down and a general overhaul carried out.

NOTICE

Important: The general overhaul must be carried out by qualified personnel in a specialised workshop with appropriate equipment in observance of applicable legal obligations.

Checking The Oil Level

Check the oil level with an oil temperature of between +20 °C...+40 °C. Extract the oil level hole plug, and check the oil level. The correct oil level is at the lower edge of the oil level hole. Top off with the correct oil type (Mobil synthetic gear oil type SHC 636 [ISO 6743 VG 680]) as necessary. Reinstall the oil level hole plug and tighten to the correct torque.

Changing The Oil

To change the oil in the gearhead, proceed as follows:

1. Place a catchment vessel under the oil drain screw.
2. Unscrew and extract the oil level hole plug.
3. Unscrew and extract the oil drain plug.
4. Drain all the oil from the gearhead.
5. Inspect the seal ring and plug, replace worn or damaged seal and plug if threads are damaged.
6. Reinstall the drain plug with seal into the drain hole and tighten to the correct torque.
7. Using a suitable filling device, refill with oil of the same type through the oil level hole (or through the pressure vent hole) until oil emerges through the oil level hole.
8. Wait at least 15 minutes and then check the oil level. Adjust oil level if necessary and reinstall the oil level hole plug to the correct torque.

The locations of the vent screw, drain plug, and oil level plug will vary by specific gearhead model and mounting orientation, reference the following graphics to find the specific arrangement for the unit under service.

⚠ WARNING

Residual burn hazard due to hot oil: Allow the gearmotor to cool down before carrying out maintenance or repair work. Wear protective gloves.

⚠ WARNING

Particles or liquids thrown up during servicing and maintenance can be injurious. Use caution when cleaning with compressed air or water. Use protective eyewear.

⚠ WARNING

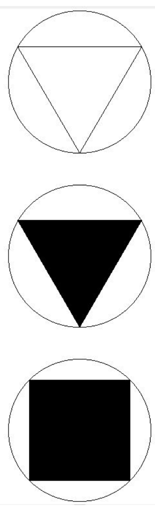
Whenever attempting to service or install this type of product, it is strongly recommended that the motor be put in a safe condition (power off) prior to servicing. Failure to follow the instructions in this manual may result in product damage, equipment damage, and serious or fatal injury to personnel.

NOTICE

Important: Take care that no dirt or water enters the shaft sealing rings or the vents when cleaning. Dirt or water in the shaft sealing rings may cause leaks.

Bolt torques [Nm]						
Dimensions	Screw connections in the strength classes			Cover screws	Threaded pin on coupling	Screw connections on protective covers
	8.8	10.9	12.9			
M4	3.2	5	6	-	-	-
M5	6.4	9	11	-	2	-
M6	11	16	19	-	-	6.4
M8	27	39	46	11	10	11
M10	53	78	91	11	17	27
M12	92	135	155	27	40	53
M16	230	335	390	35	-	92
M20	460	660	770	-	-	230
M24	790	1150	1300	80	-	460
M30	1600	2250	2650	170	-	-
M36	2780	3910	4710	-	-	1600
M42	4470	6290	7540	-	-	-
M48	6140	8640	16610	-	-	-
M56	9840	13850	24130	-	-	-
G½	-	-	-	75	-	-
G¾	-	-	-	110	-	-
G1	-	-	-	190	-	-
G1¼	-	-	-	240	-	-
G1½				300		-

Oil Plug Type Locator Drawings

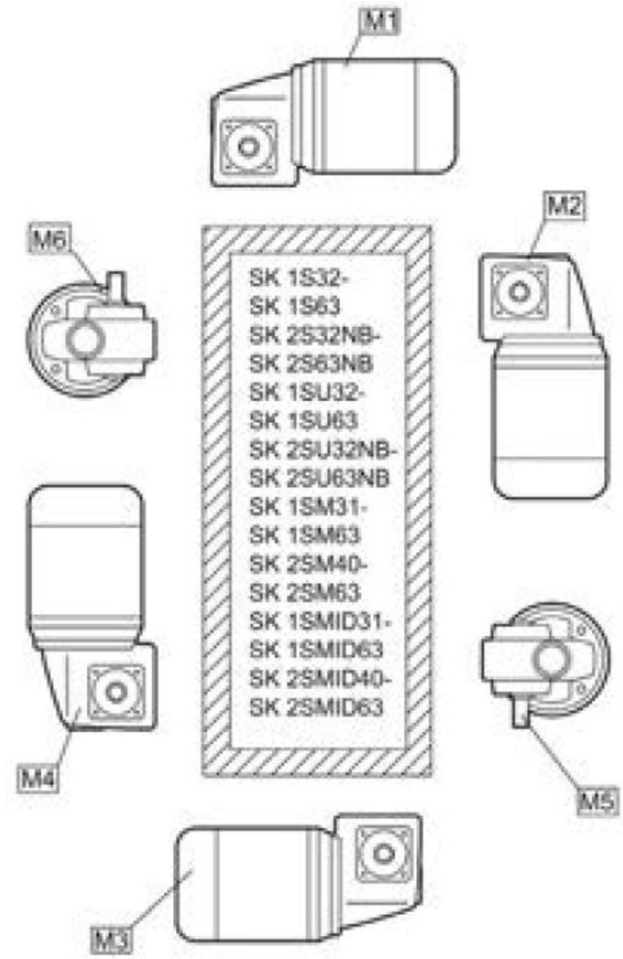


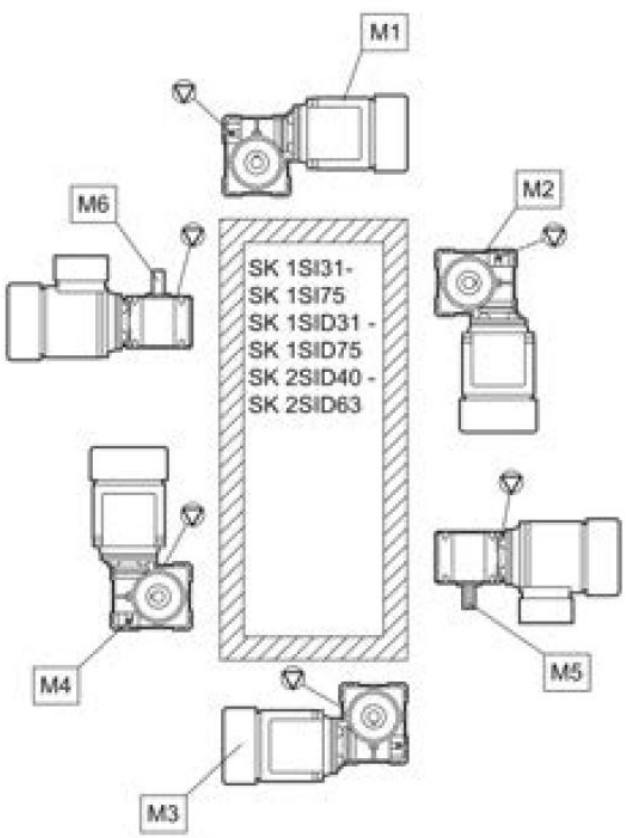
Vent Screw

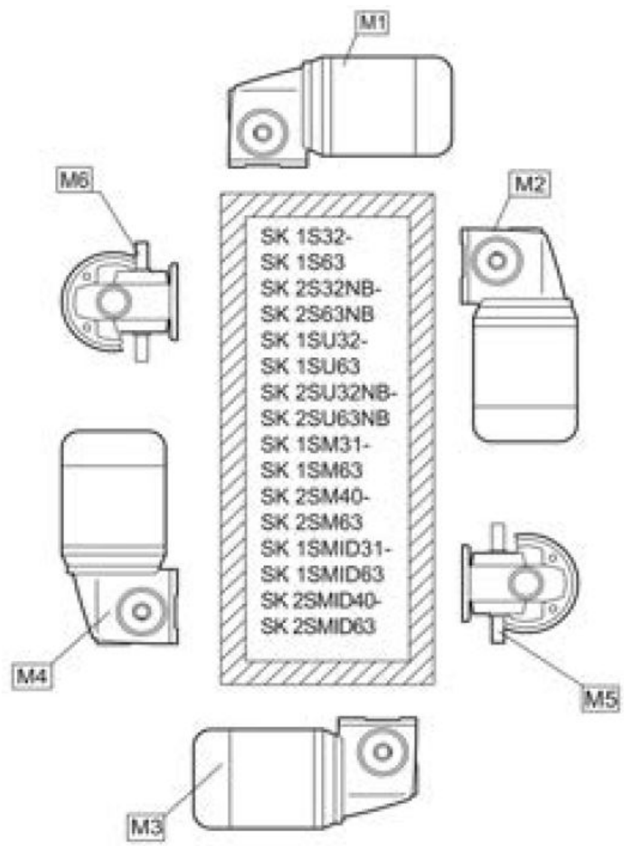
Oil Level Hole Plug

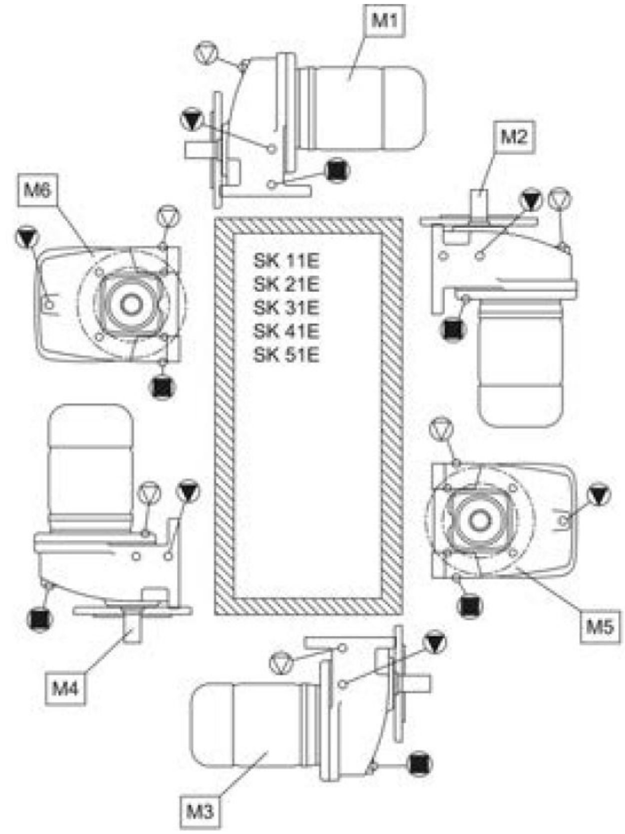
Oil Drain Plug

Plug Type Legend





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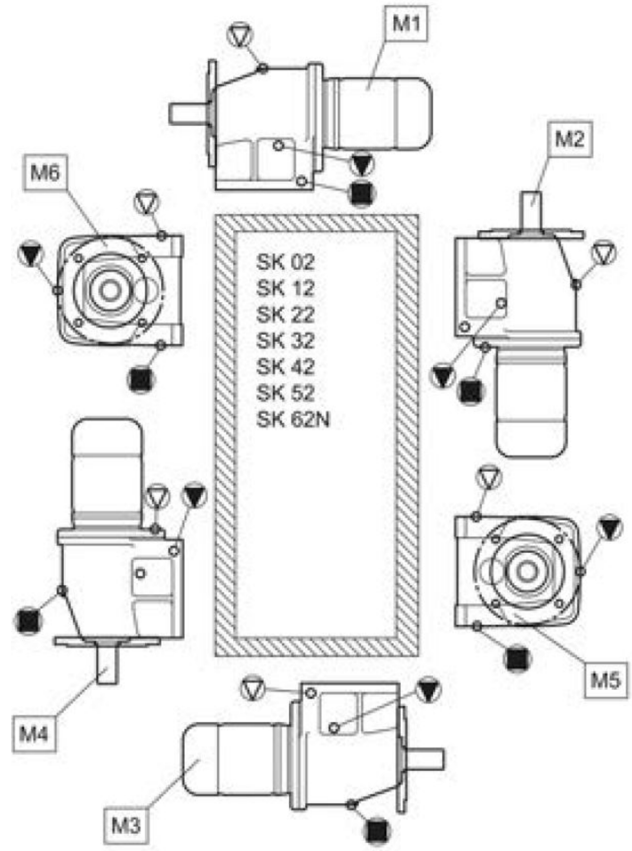
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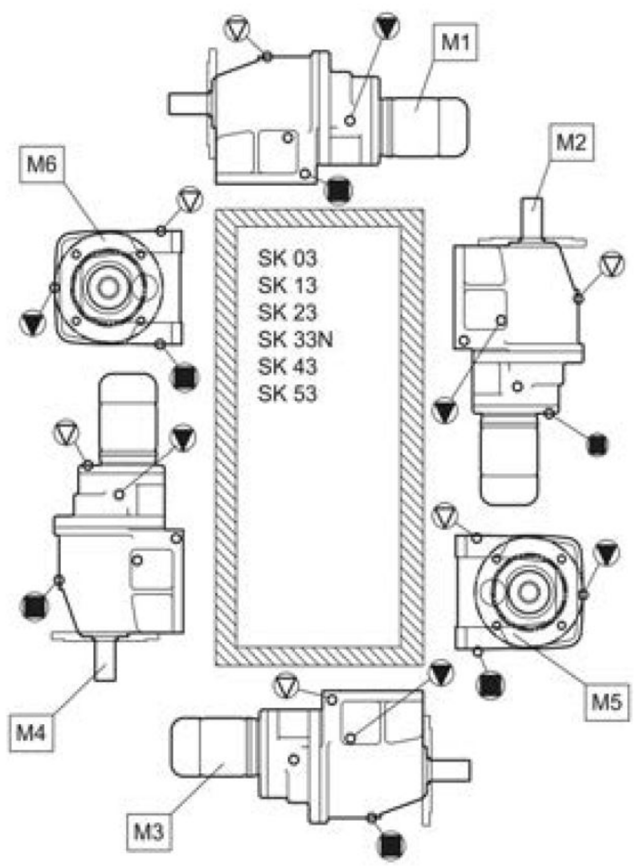
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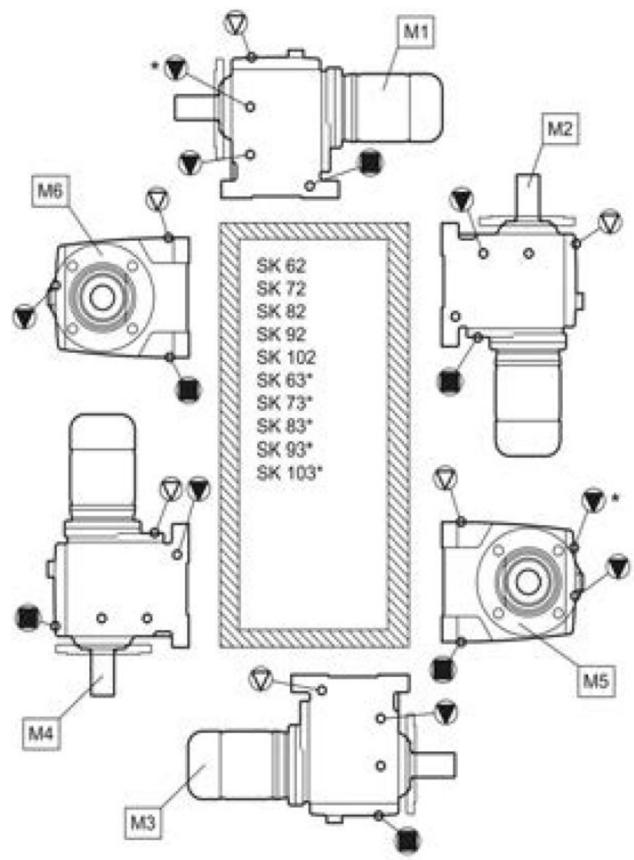
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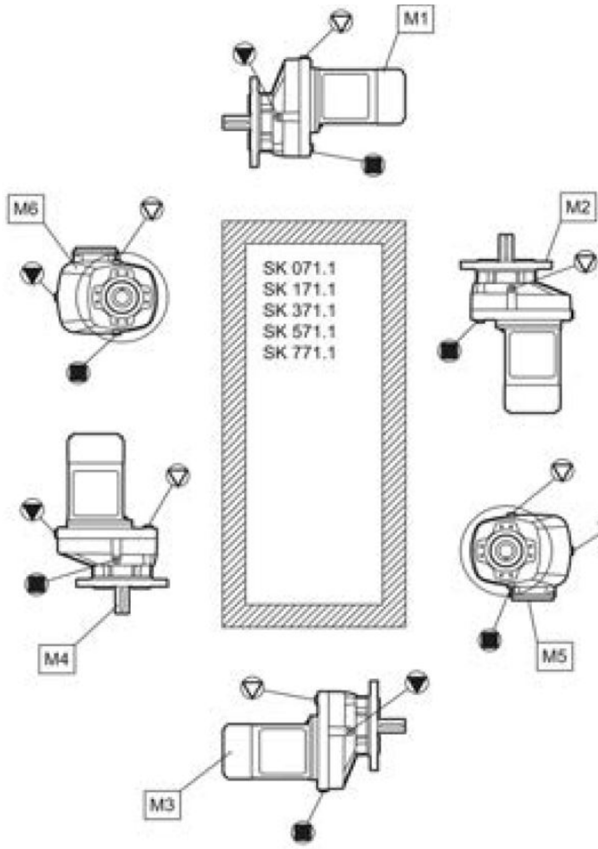
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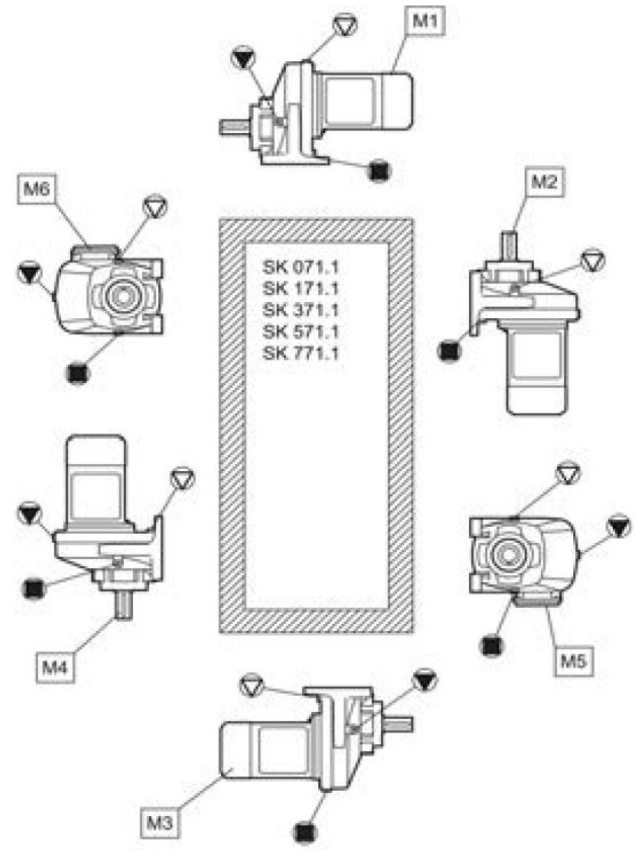
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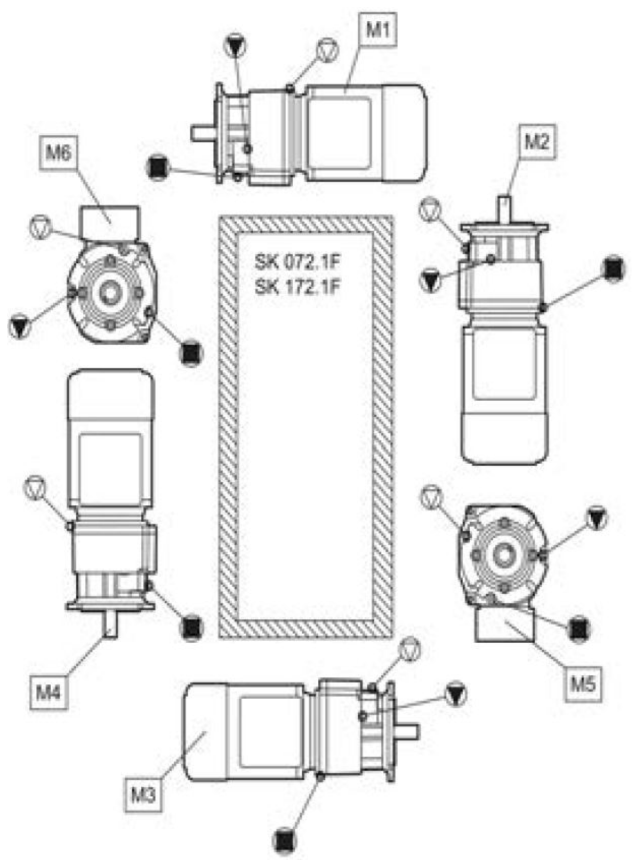
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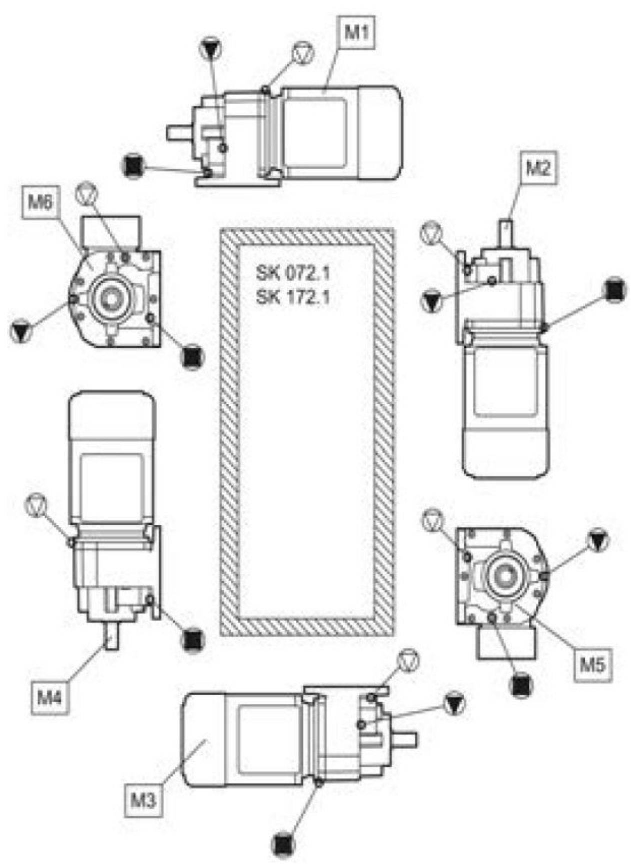
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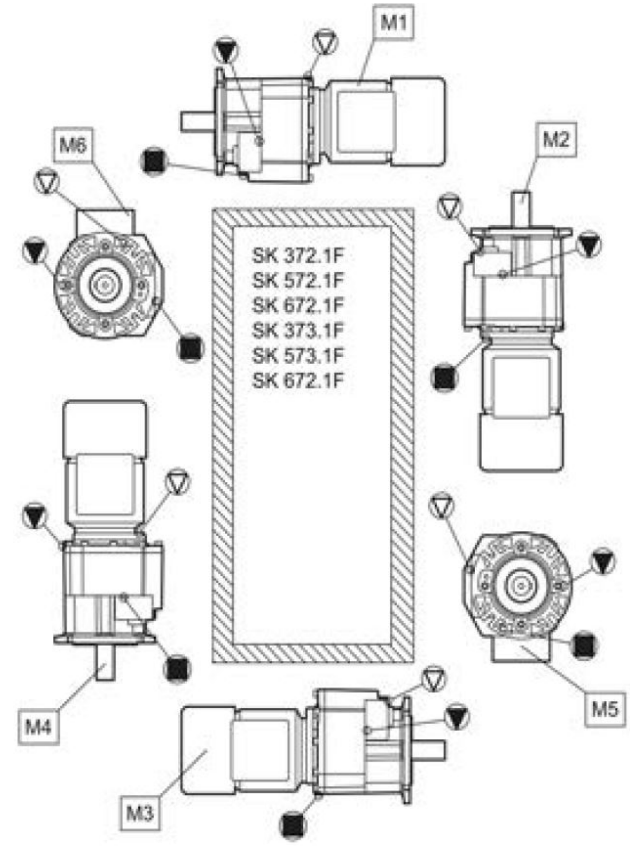
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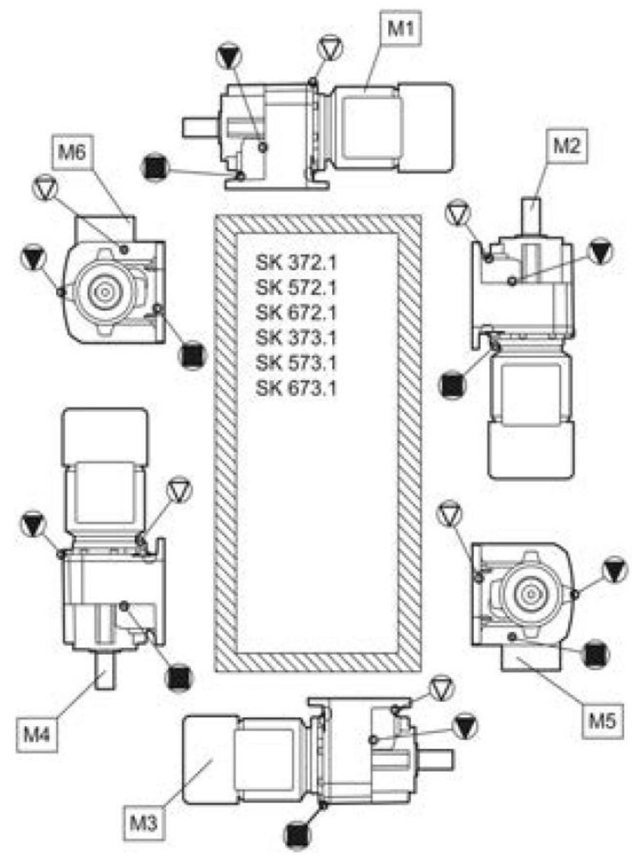
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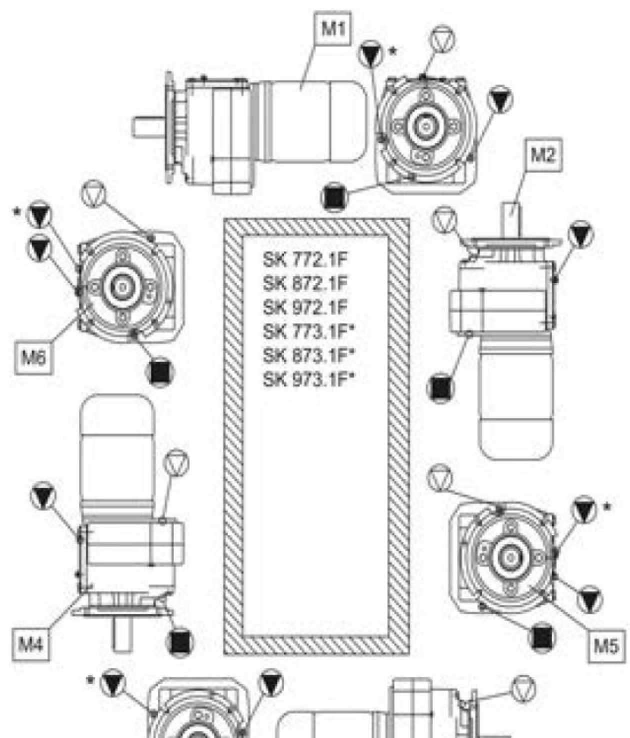
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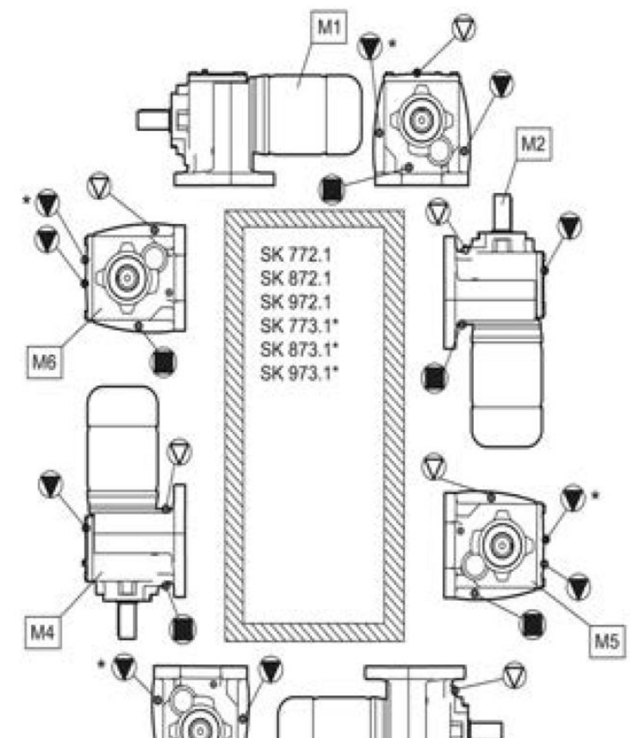
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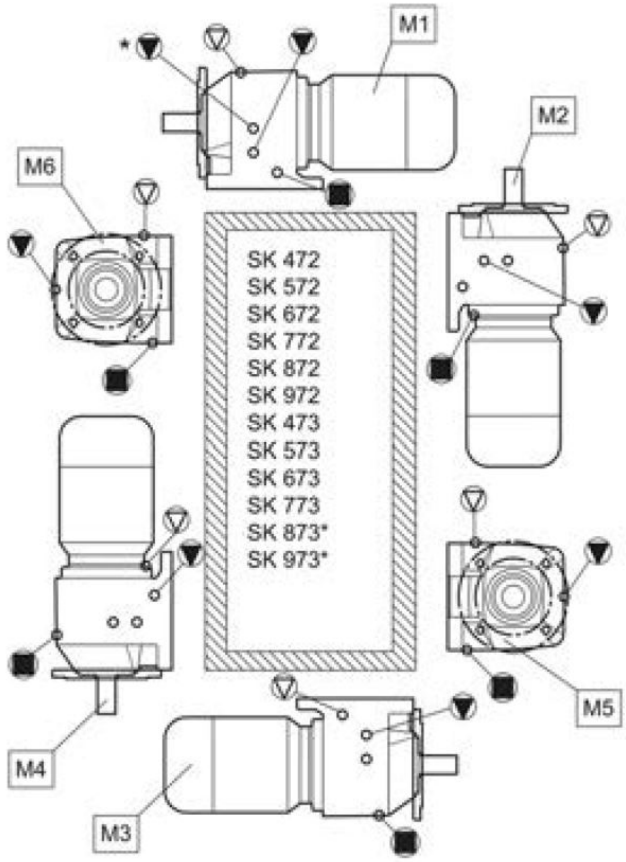
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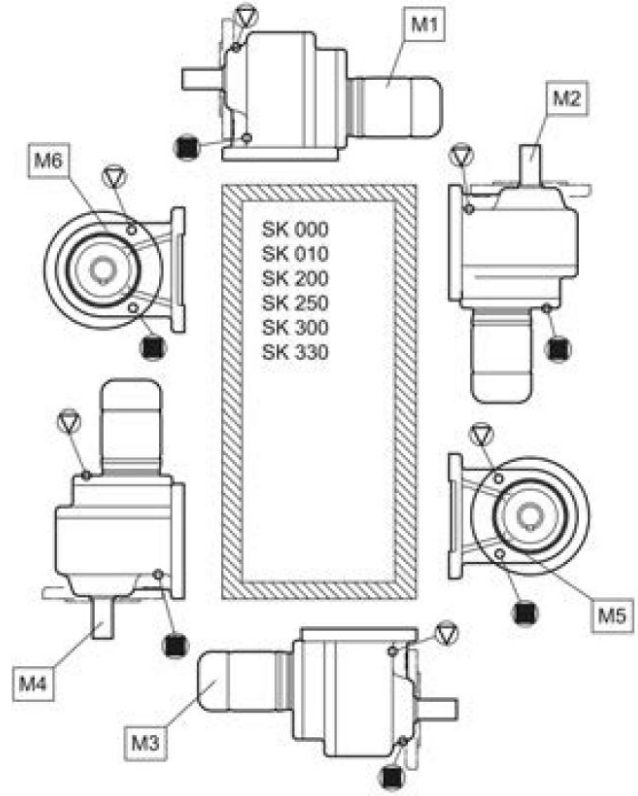
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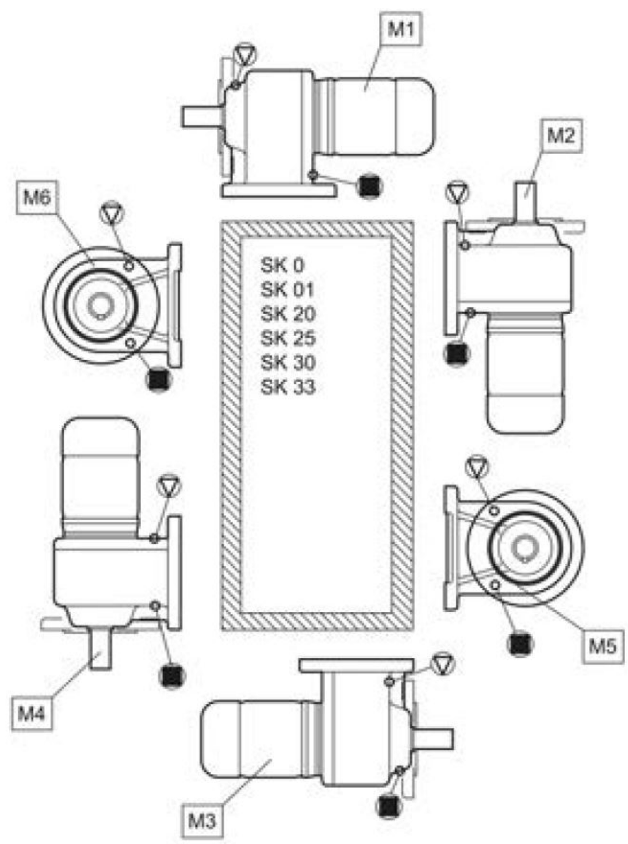
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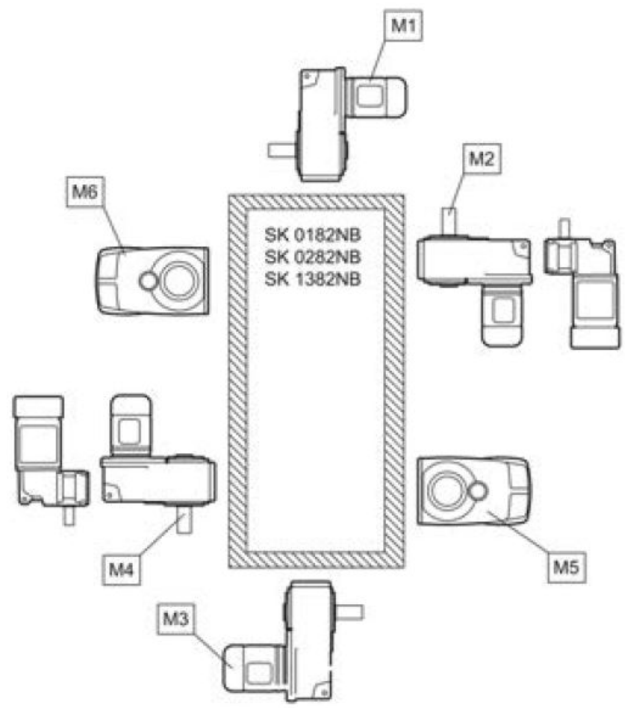
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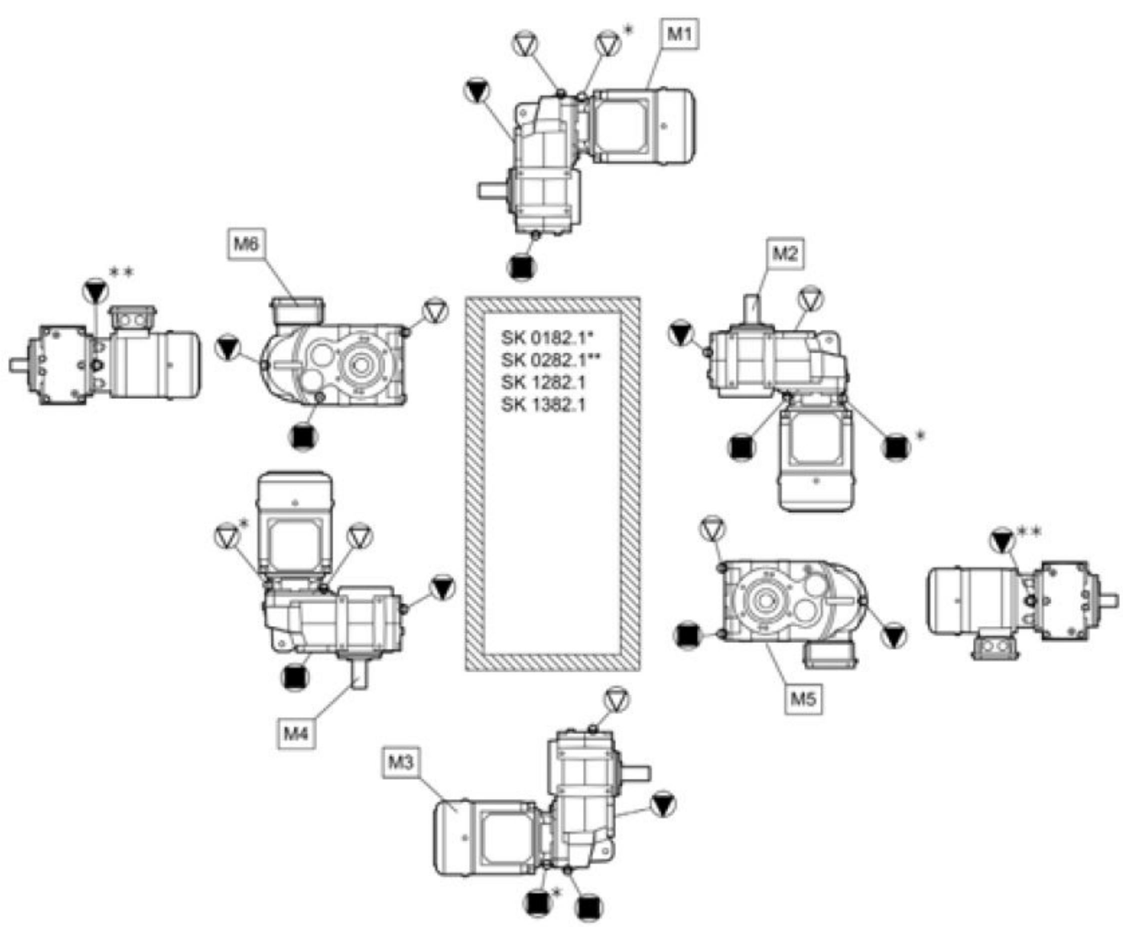
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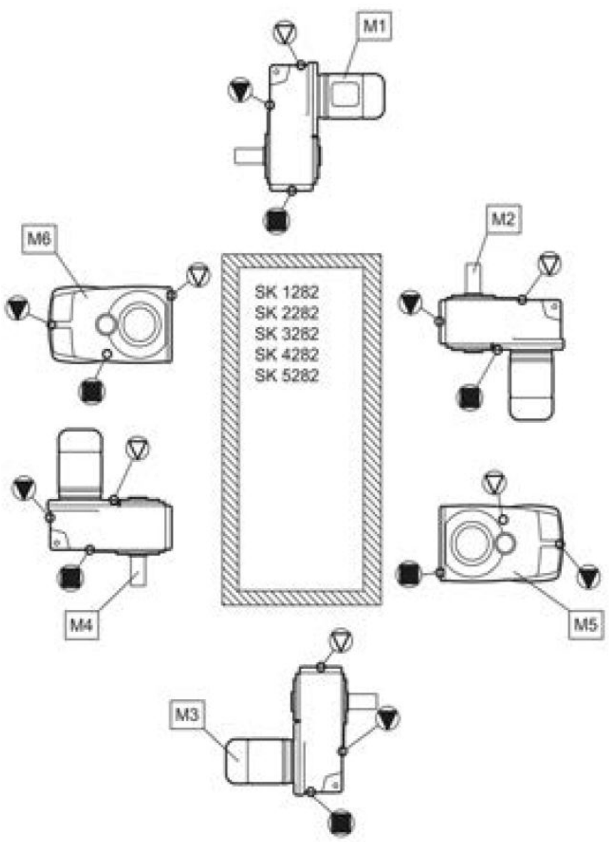
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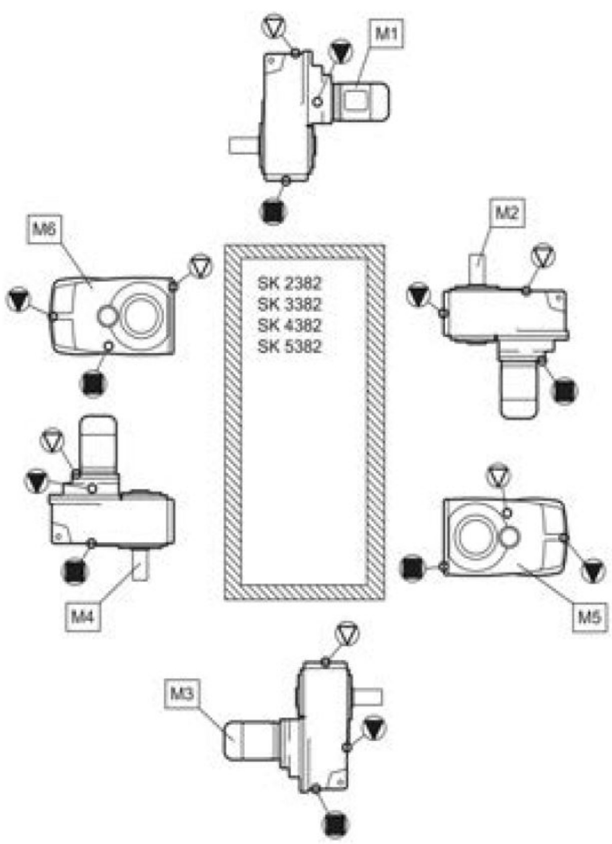
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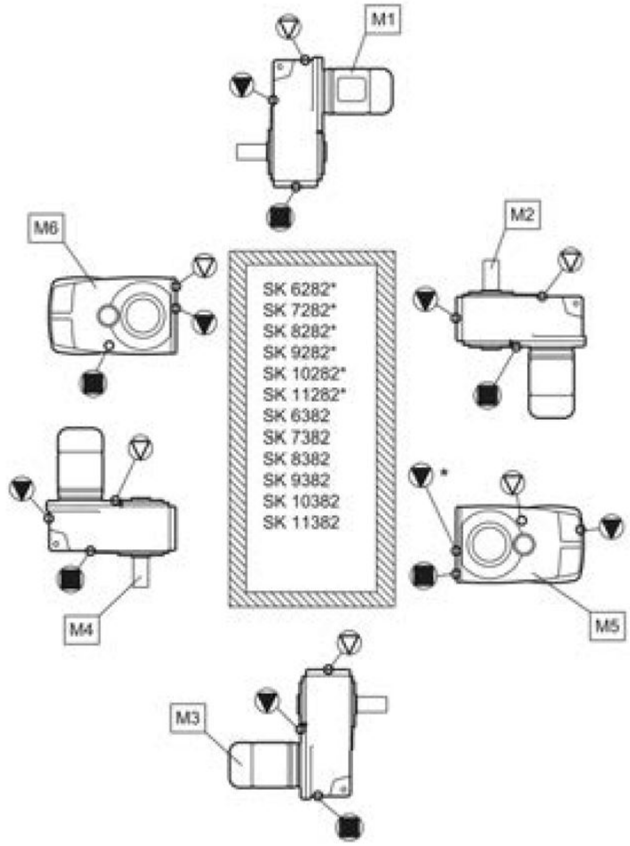
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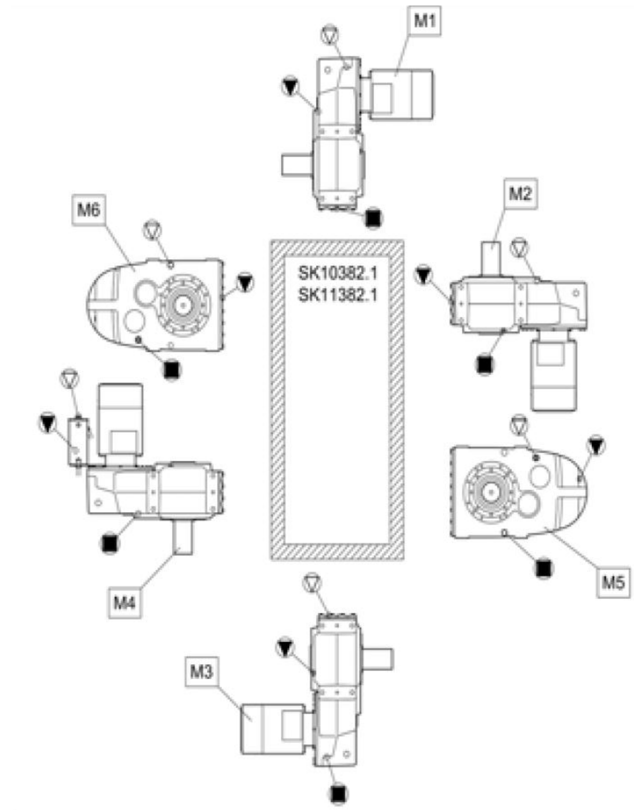
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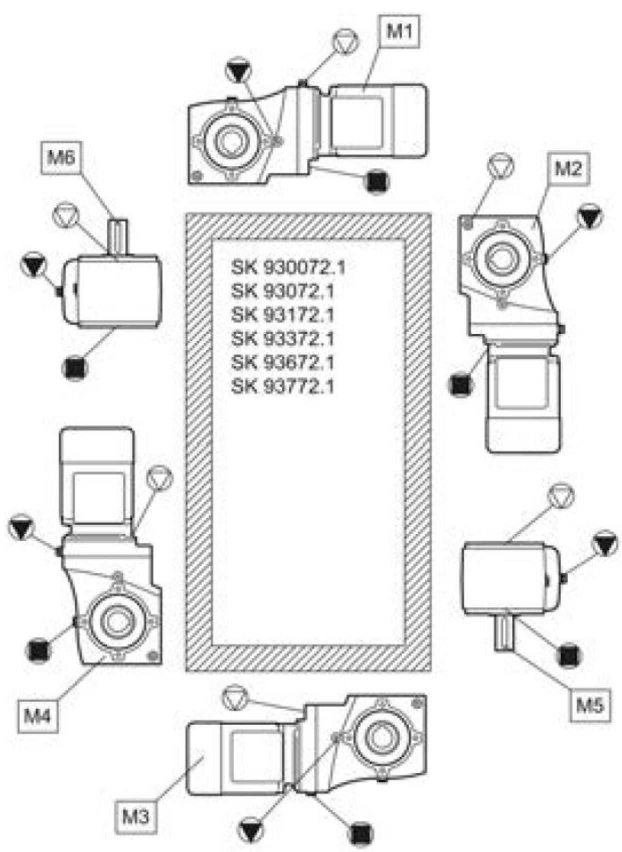
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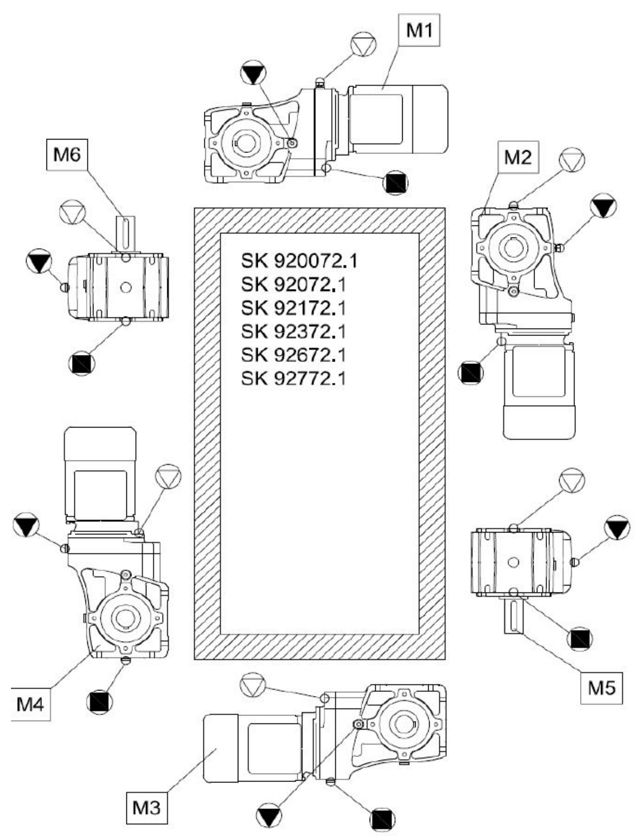
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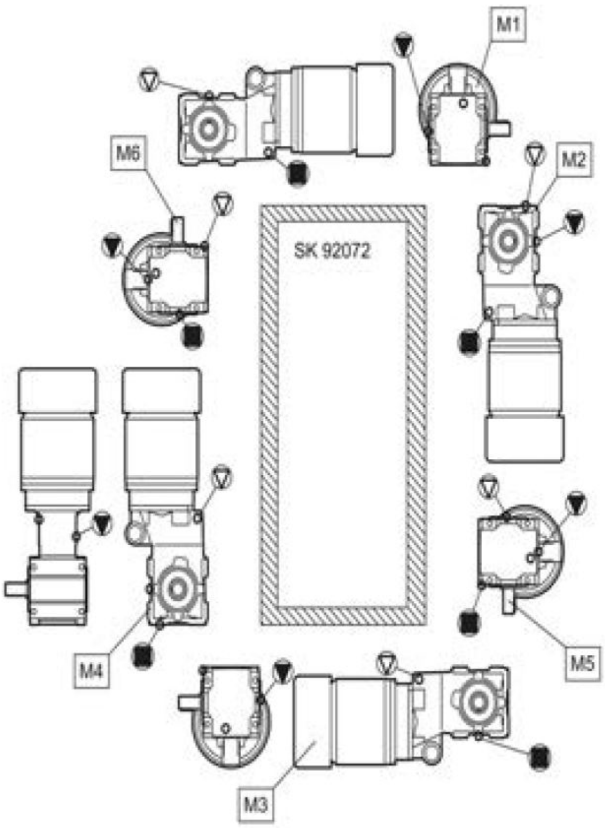
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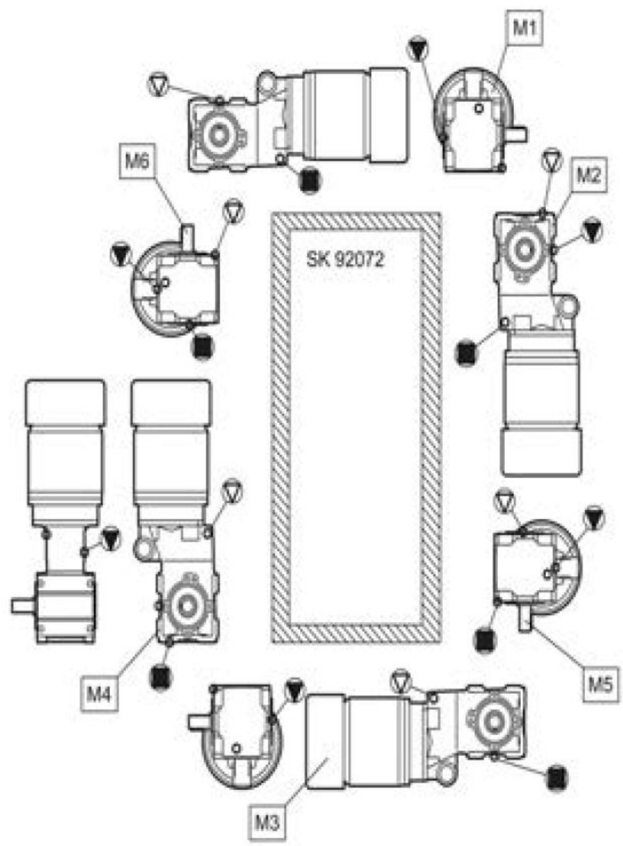
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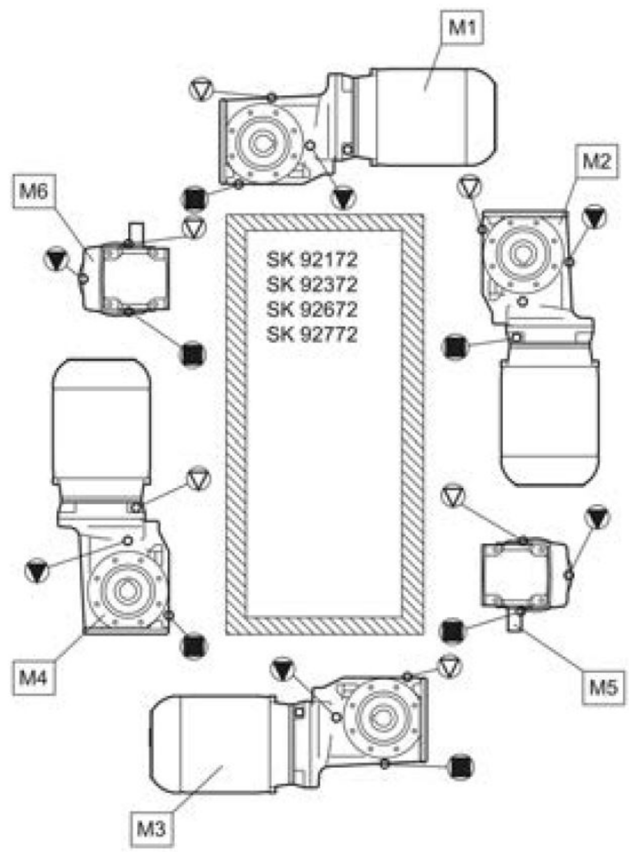
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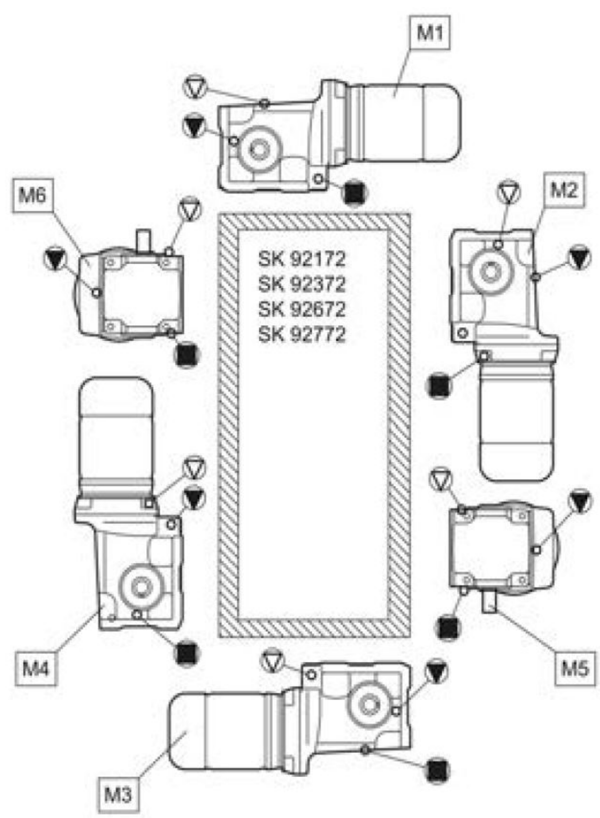
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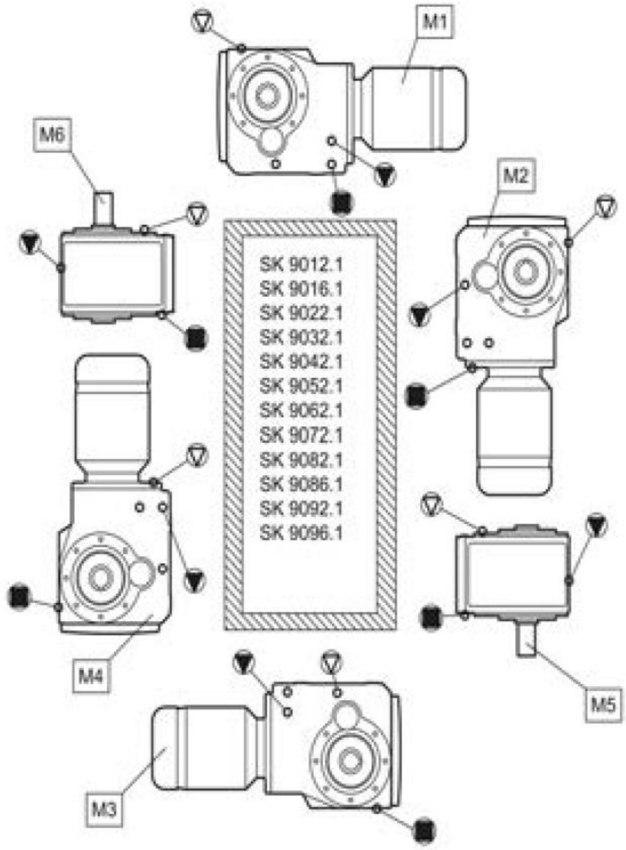
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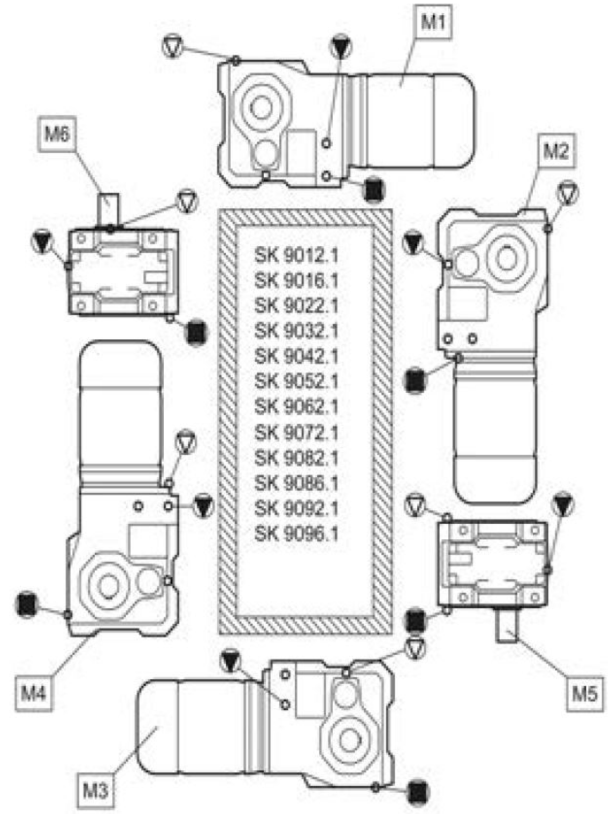
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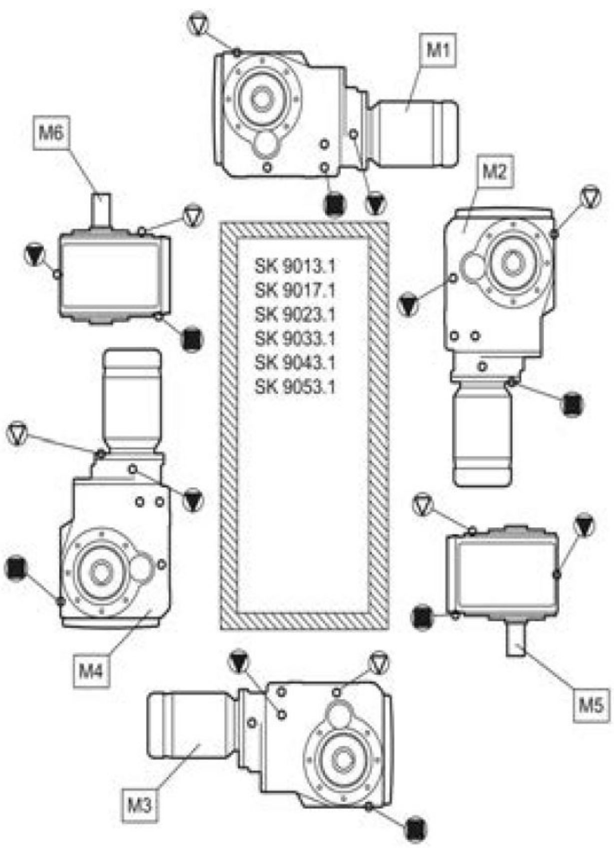
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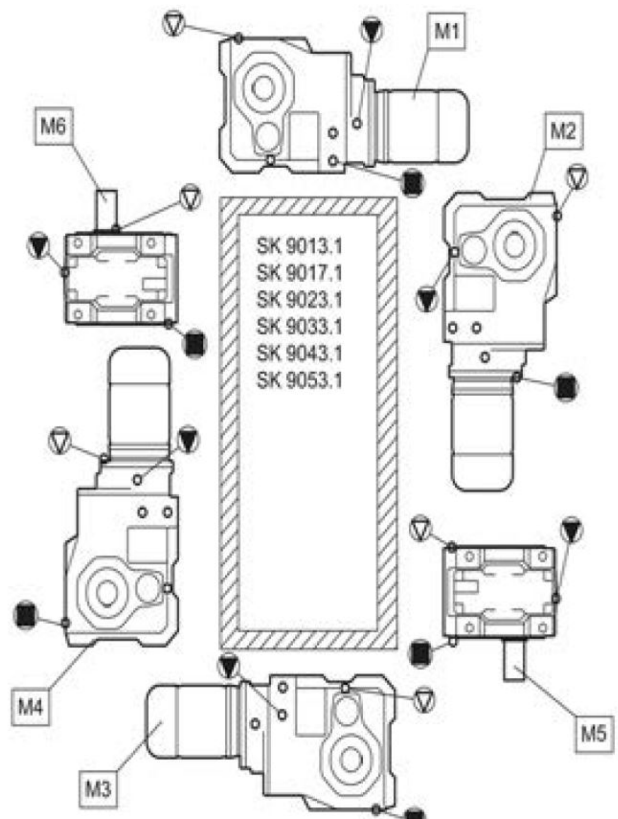
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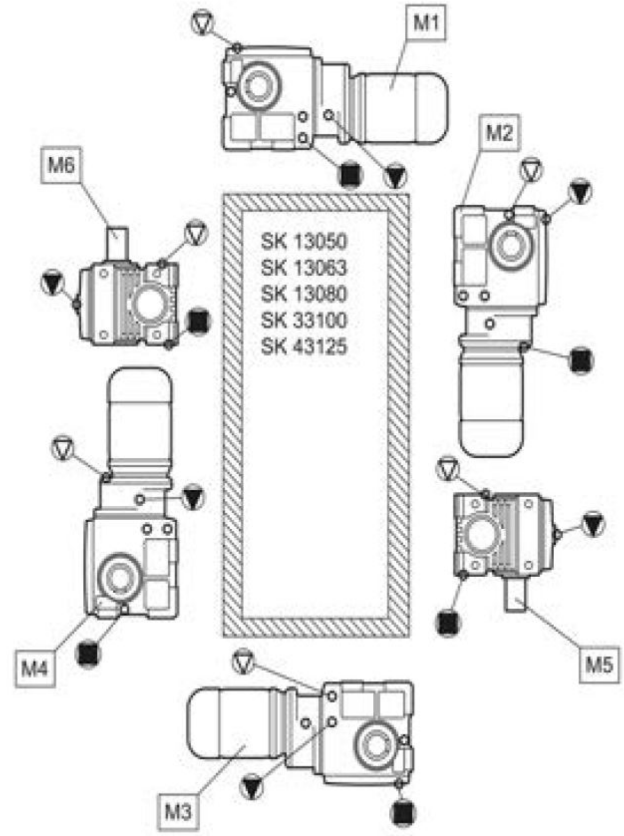
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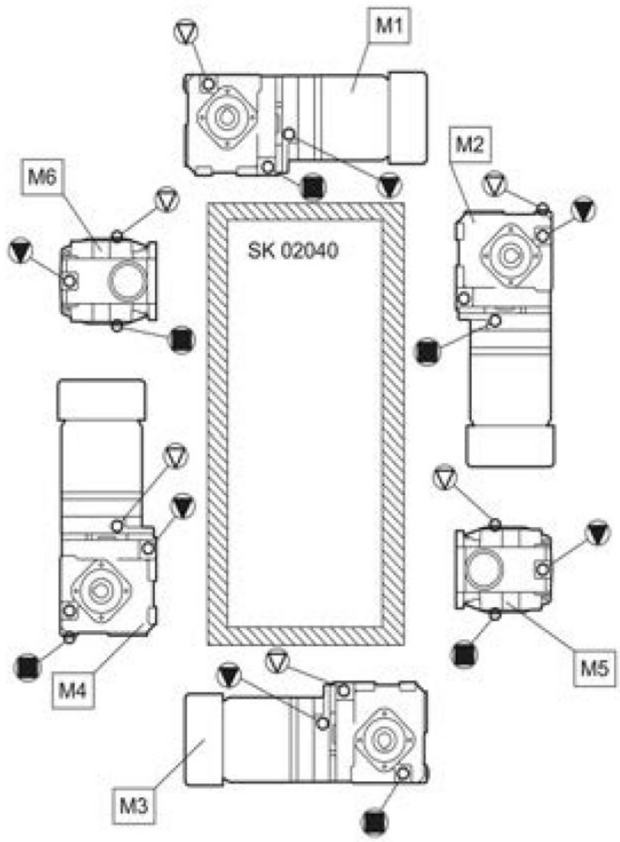
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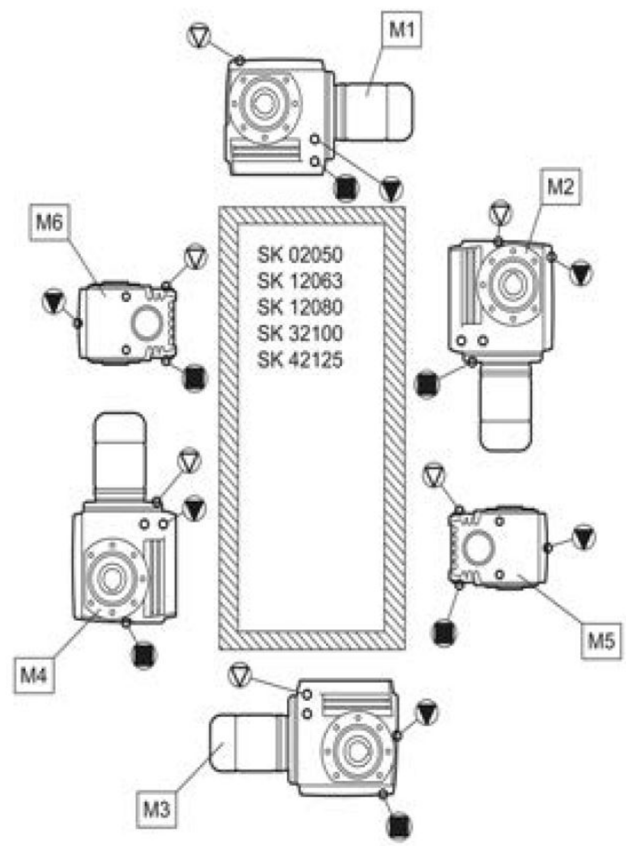
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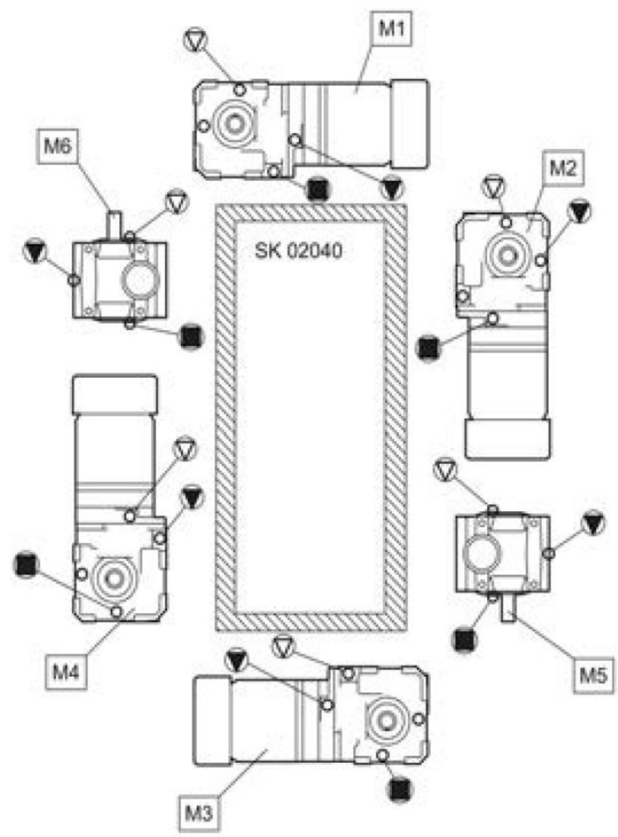
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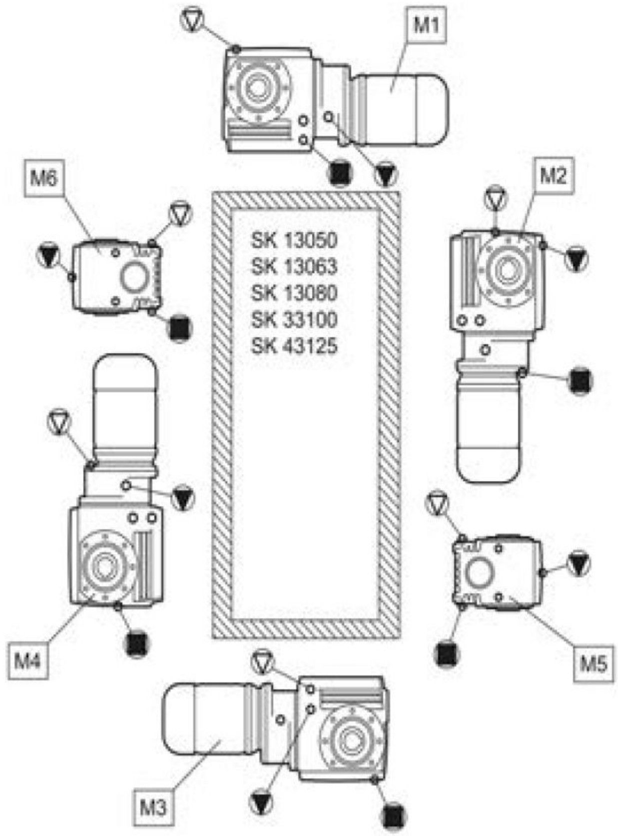
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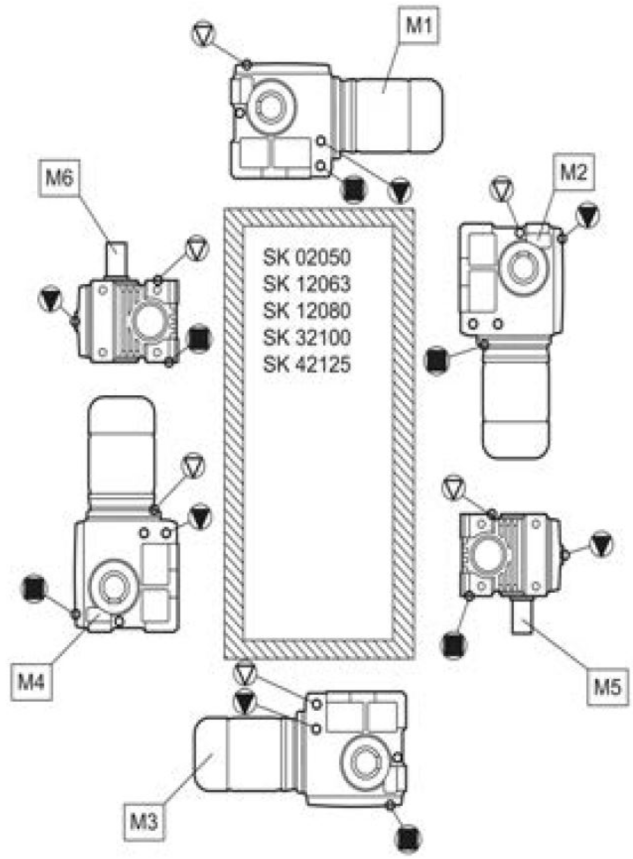
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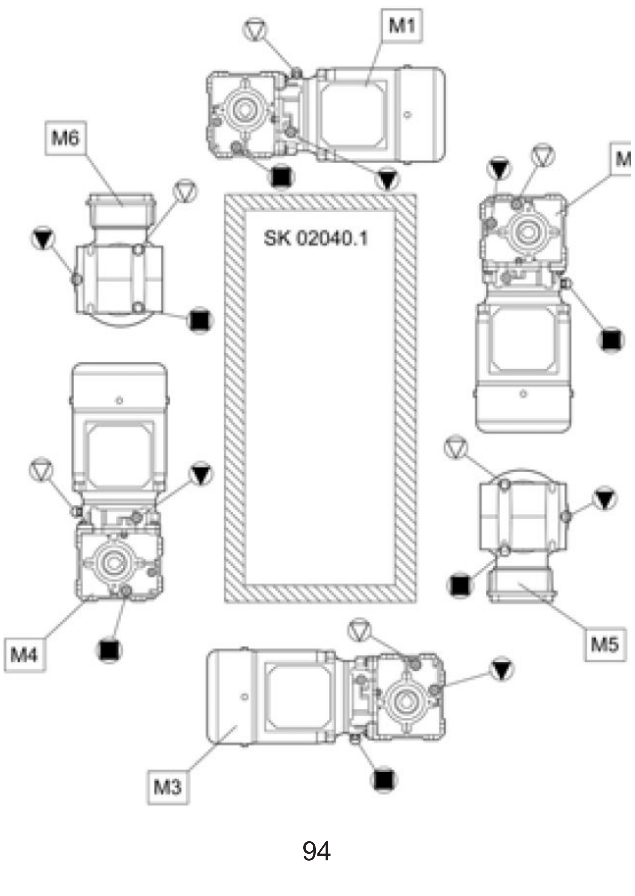
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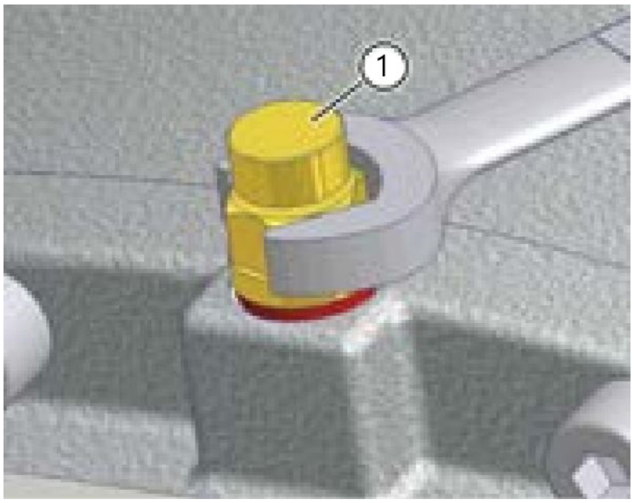
Relubricating the Bearings

For bearings which are not oil-lubricated, and whose holes are completely above the oil level, replace the roller bearing grease (recommended grease: Klüber PETAMO GHY 133 N).

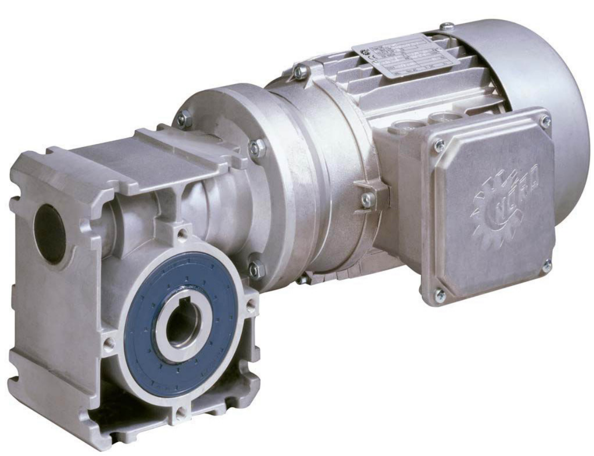
Cleaning or Replacing the Vent Screw

Unscrew the vent screw (**1**) and thoroughly clean (e.g. with compressed air) and fit the vent screw in the same place, if necessary use a new vent screw with a new seal ring.



Replacing the Shaft Sealing Ring

Once the shaft sealing ring has reached the end of its service life, the oil film in the region of the sealing lip increases and a measurable leakage with dripping occurs. The shaft sealing ring must then be replaced. The space between the sealing lip and the protective lip must be filled approximately 50% with grease on fitting . Take care that after fitting (recommended grease: Klüber PETAMO GHY 133 N), the new shaft sealing ring does not run in the old wear track.



Asynchronous Gearmotor

Gearhead Replacement

When replacing defective gearheads it is advisable to consult the installation instruction bulletin furnished with the new Gearhead by the OEM.

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